London Tube

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Data Source: https://github.com/jaron/railgraph

Introduction

The "London Underground Network" refers to the metro system of London, UK, encompassing its stations and the connections between them. The network consists of 309 vertices (representing stations), and 370 edges (representing the metro line connections between stations). Notably, multiple metro lines can connect to the same station, resulting in multiple edges. As an undirected network, each edge signifies a bidirectional connection, meaning travel is possible in both directions between stations.

Data loading and cleanning

The following initial part will be for loading the Network and enriching the nodes with usage data from a seperate dataset.

In [1]:
```
# Load Libraries
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import random

import contextily as ctx
import networkx as nx
import igraph as ig

from networkx.algorithms.community import greedy_modularity_communi
from networkx.algorithms.community.quality import modularity

from networkx.algorithms.community import label_propagation_communi
from networkx.algorithms.community import girvan_newman

import community.community_louvain as community_louvain
```

Loading the Network and extracting the labels of the Stations, as they are going to be used as a way to named the vertex of the network.

In [2]:
```
#Read the Network and extract geographic positions
G = nx.read_graphml('data/london_tube/tubeDLR.graphml')
labels = nx.get_node_attributes(G, 'label')
pos_geo = {node: (float(data['longitude']), float(data['latitude'])
```

Remove node attributes that were delivered with the network.

In [3]:
```
# Remove unwanted attributes
attributes_to_remove = [
    'Betweenness Centrality', 'Closeness Centrality', 'Clustering C
    'Degree', 'Eccentricity', 'Eigenvector Centrality', 'In-Degree'
    'Number of triangles', 'Out-Degree', 'PageRank', 'b','g',
    'latitude_copy', 'longitude_copy', 'r', 'size', 'x', 'y','displ
]

for node, data in G.nodes(data=True):
    for attr in attributes_to_remove:
        data.pop(attr, None)
```

To further enrich the "London Underground Network", a seperate Dataset is taken, it contains the average entry and exit values from most of the stations (vertex) in our Network and add theses to our nodes. To do so, we first remove all stations where no numbers are available, which turns out to corresponds to the Docklands Light Railway (DLR) and thus can be knowingly discarded.

In [4]:
```
### Add Usage Data
usage_data = pd.read_csv('data/london_tube/2017_Entry_Exit.csv')

# Feature data modification
usage_data['Avg_Exit_Weekend'] = usage_data[['Exit_Saturday', 'Exit_
usage_data['Avg_Entry_Weekend'] = usage_data[['Entry_Saturday', 'En

usage_data.drop(columns=['Entry_Saturday', 'Entry_Sunday', 'Exit_Sa

# Define mapping from non-standard to standard names
station_name_corrections = {
    'Edgware Road (Cir)': 'Edgware Road',
    'Kew Gardens': 'Kew Gardens (London)',
    "Shepherd's Bush (H&C)": "Shepherd's Bush Market",
    "Shepherd's Bush (Cen)": "Shepherd's Bush",
    'Hammersmith (Dis)': 'Hammersmith (Piccadilly and District line
    'Hammersmith (H&C)': 'Hammersmith (Hammersmith & City and Circl
    "St. John's Wood": "St John's Wood",
    'Bank & Monument': 'Bank-Monument',
    'Totteridge & Whetstone': 'Totteridge and Whetstone',
    'Paddington': 'London Paddington',
    'Richmond': 'Richmond (London)',
    'Heathrow Terminals 123': 'Heathrow Terminals 1, 2, 3',
    'Edgware Road (Bak)': 'Edgware Road (Bakerloo line)',
    'Victoria': 'London Victoria'
}

# Apply corrections to df.Stations
usage_data['Station'] = usage_data['Station'].replace(station_name_
```

```
# Filter nodes in the graph that are available in usage_data
stations_in_usage_data = set(usage_data['Station'])
nodes_to_keep = [node for node, data in G.nodes(data=True) if data.

# Identify stations to be dropped
stations_to_drop = [labels[node] for node, data in G.nodes(data=Tru


# Create a subgraph with only the nodes to keep
G = G.subgraph(nodes_to_keep).copy()

# Add Entry_Week, Exit_Week, Avg_Exit_Weekend, Avg_Entry_Weekend to
for node, data in G.nodes(data=True):
    station_label = data.get('label')
    if station_label in stations_in_usage_data:
        station_data = usage_data[usage_data['Station'] == station_
        data['Entry_Week'] = station_data['Entry_Week']
        data['Exit_Week'] = station_data['Exit_Week']
        data['Avg_Exit_Weekend'] = station_data['Avg_Exit_Weekend']
        data['Avg_Entry_Weekend'] = station_data['Avg_Entry_Weekend

        # Remove edges connected to nodes that are not in the filte
        G.remove_edges_from([(u, v) for u, v in G.edges() if u not
```

The Euclidean Distance between Nodes as a attribute of the arcs using longitude/latitude of the nodes.

In [5]:
```
# Add "dist" attribute to each edge based on the euclidean distance
for u, v, data in G.edges(data=True):
    lon1, lat1 = pos_geo[u]
    lon2, lat2 = pos_geo[v]
    distance = np.sqrt((lon2 - lon1)**2 + (lat2 - lat1)**2) * 111
    data['dist'] = distance
```

Finally , the working "London Underground Network" is ready to be used for the upcoming dofferent analysis. Here it is listed the nodes (vertex) and edges (links) attributes.

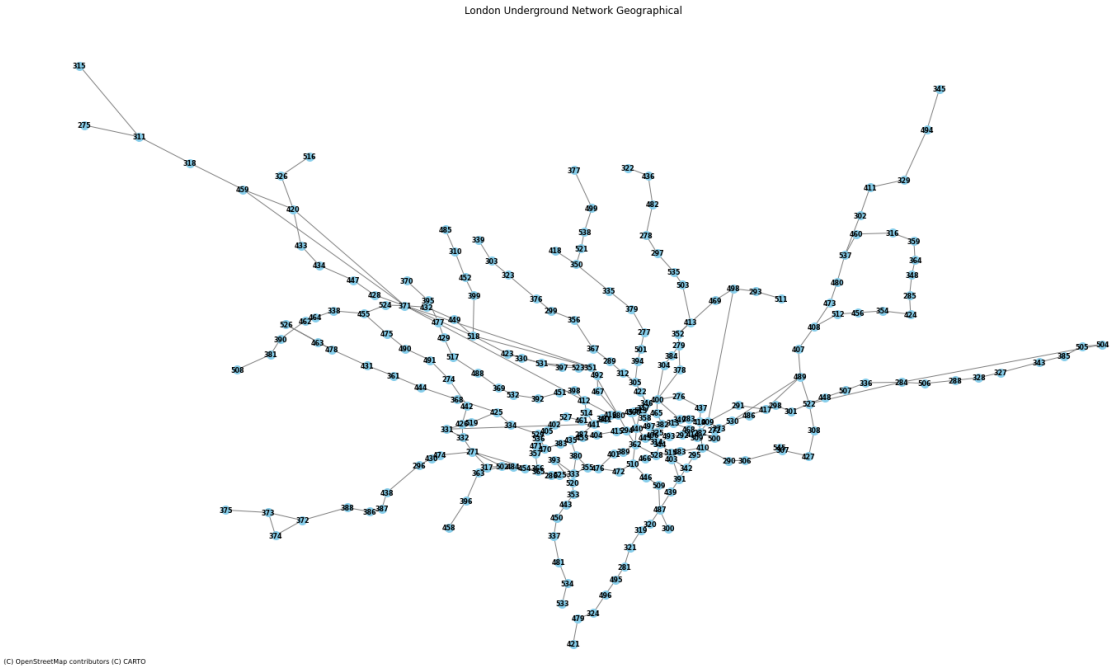
In [6]:
```
# List Attributes

# Nodes
unique_keys = set().union(*(data.keys() for _, data in G.nodes(data=
print("Unique Node Attributes:")
for key in sorted(unique_keys):
    print(f" - {key}")

# Edges
unique_edge_keys = set().union(*(data.keys() for _, _, data in G.edg
print("\nUnique Edge Attributes:")
for key in sorted(unique_edge_keys):
    print(f" - {key}")
```

```
Unique Node Attributes:
 - Avg_Entry_Weekend
 - Avg_Exit_Weekend
 - Entry_Week
 - Exit_Week
 - displayName
 - label
 - latitude
 - longitude
 - stationReference

Unique Edge Attributes:
 - Neo4j Relationship Type
 - dist
 - line
 - weight
```

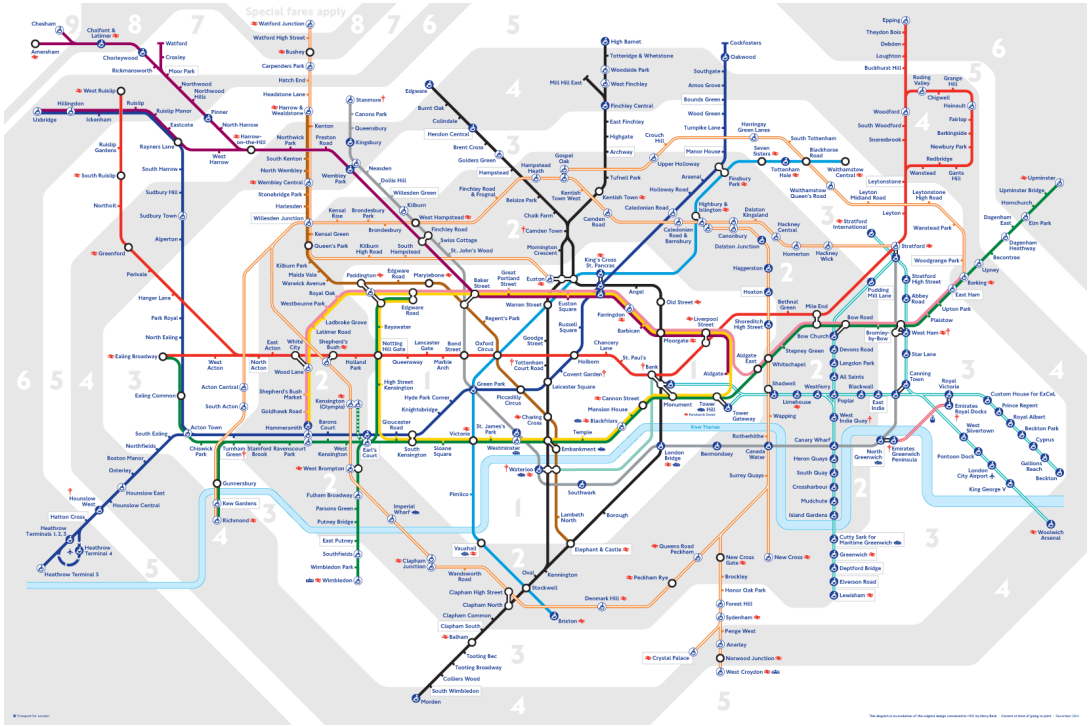
Network properties

The basic properties of the "London Underground Network" are analysed to get a general idea of the characteristics of it. The initial step is to plot the network with the node position corresponding to the longitude and latitude values into a map of the City of London to get a general feel for the geography underlying the network. After plotting visualy the grahp, its clear how the Network consist out of arms corresponding to the connection between nodes and egdes in the outside area of the city center, also its highligted a concentration of nodes, as they are metro stations, where changes between lines are possible. Therefore, we are in the presence of a transportation Network.

In [7]:
```
# Plot network with map of the city of London
fig, ax = plt.subplots(figsize=(25, 25))

ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

nx.draw(G, pos=pos_geo, with_labels=True, node_size=100, node_color
        font_size=8, font_weight='bold', edge_color='gray', ax=ax)

# Cosmetics and plot
ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network Geographical")
plt.show()
```



It can be compared the "London Underground Network" to an official map of the City of London Transportation System, here its reinforced the idea of the same arm and central hubs structure, that can be denominated as tree structure, with the metro stations as vertex that are connected throught the metro lines as edges.



The "London Underground Network", after the data cleaning, can be represented as an undirected network with 269 nodes (stations) and 324 edges (connections between stations).

In [8]:
```
# Basic network characteristics
if isinstance(G, nx.DiGraph):
    print("The graph is directed.")
```

```
else:
    print("The graph is undirected.")

print("Number of Nodes (Order):", G.number_of_nodes())
print("Number of Edges:", len(G.edges(data=True)))
```

```
The graph is undirected.
Number of Nodes (Order): 269
Number of Edges: 324
```

Then, it is checked if there are any isolated vertices (stations) in the network. It is found the isolated vertex '545' in the London Underground network, which refers to a station that has no connections to any other stations, making it unreachable within the system. In this case, the isolated station is produced as a error of the initial enrichment of with usage data wich left the node isolated. Therefote, the node is deleted to correct for that error.

In [9]:
```
# Find isolated vertices
print("Isolated Vertices: ",list(nx.isolates(G)))
print(labels["545"])

# Remove node '545' from labels and the graph
labels.pop("545", None)
G.remove_node("545")

print("Isolated Vertices: ",list(nx.isolates(G)))
```

```
Isolated Vertices:  ['545']
Canary Wharf
Isolated Vertices:  []
```

In network analysis, specifically in the context of the "London Underground Network", it's important to identify adjacent vertices (stations) and edges (metro lines). This helps to understand the connections between stations and the metro lines serving them. The following code allows us to find these adjacent vertices, revealing the connections between different metro lines at a given station. Additionally, the adjacent edges provide information about possible line changes that a station can offer, indicating where passengers can switch between different metro lines.

In [10]:
```
# Get the adjacent vertices
chosen_vertex = '487'

adjacent_vertices = list(G.neighbors(chosen_vertex))
print(f"Adjacent vertices of vertex {chosen_vertex}: {adjacent_vert

adjacent_labels = [labels[vertex] for vertex in adjacent_vertices]
print(f"Adjacent vertices of vertex {labels[chosen_vertex]}: {adjac
```

```
Adjacent vertices of vertex 487: ['300', '320', '439', '509']
Adjacent vertices of vertex Stockwell: ['Brixton', 'Clapham North',
'Oval', 'Vauxhall']
```

In [11]:
```
# Get the adjacent edges
adjacent_edges = G.edges(chosen_vertex)
print(f"Adjacent edges of vertex {chosen_vertex}: {adjacent_edges}"

for edge in adjacent_edges:
    line_name = G[edge[0]][edge[1]].get('line', 'Unknown Line')
    print(f"Edge: {edge} (Connecting {labels[edge[0]]} and {labels[
```

```
Adjacent edges of vertex 487: [('487', '300'), ('487', '320'), ('48
7', '439'), ('487', '509')]
Edge: ('487', '300') (Connecting Stockwell and Brixton) - Metro Lin
e: Victoria_line
Edge: ('487', '320') (Connecting Stockwell and Clapham North) - Metr
o Line: Northern_line
Edge: ('487', '439') (Connecting Stockwell and Oval) - Metro Line: N
orthern_line
Edge: ('487', '509') (Connecting Stockwell and Vauxhall) - Metro Lin
e: Victoria_line
```

The presence of self-loops in the network was examined, and it was determined that the system does not contain any self-loops. This finding is consistent with the design of most metro systems, where a station typically does not have a direct connection to itself via a metro line. Self-loops are generally avoided in urban transit networks for practical reasons, as they would not contribute to effective transportation or improve connectivity.

In [12]:
```
# Find self loops
self_loops = [edge for edge in G.edges() if edge[0] == edge[1]]

if self_loops:
    print(f"The graph has {len(self_loops)} self-loop(s):")
    for loop in self_loops:
        print(loop)
else:
    print("The graph does not have any self-loops.")
```

```
The graph does not have any self-loops.
```

Similarly, it was determined that the network does not contain any multi-edges, which means there are no multiple metro lines connecting the same pair of stations. This is consistent with the design principles of the metro system, where each pair of stations typically has a unique connection served by a single metro line. Having multiple lines between the same two stations is not a common practice, as it can lead to inefficiencies in the network design and complicate operations.

In [13]:
```
# # Find Multi Edges
if isinstance(G, nx.MultiGraph) or isinstance(G, nx.MultiDiGraph):
    multi_edges = [edge for edge in G.edges()]

    if multi_edges:
        print(f"The graph has multi-edges between the following nod
        for edge in multi_edges:
```

```
            print(edge)
    else:
        print("The graph does not have any multi-edges.")
else:
    print("The graph is not a MultiGraph or MultiDiGraph, so multi-
```

```
The graph is not a MultiGraph or MultiDiGraph, so multi-edges are no
t allowed.
```

Subnetworks

After getting the "London Underground Network" plot, the exploration of subnetworks and components is performed. From the official transportation map shown above, it can further be derived some of the main subnetworks corresponding to individual metro, light rail or regional train lines.

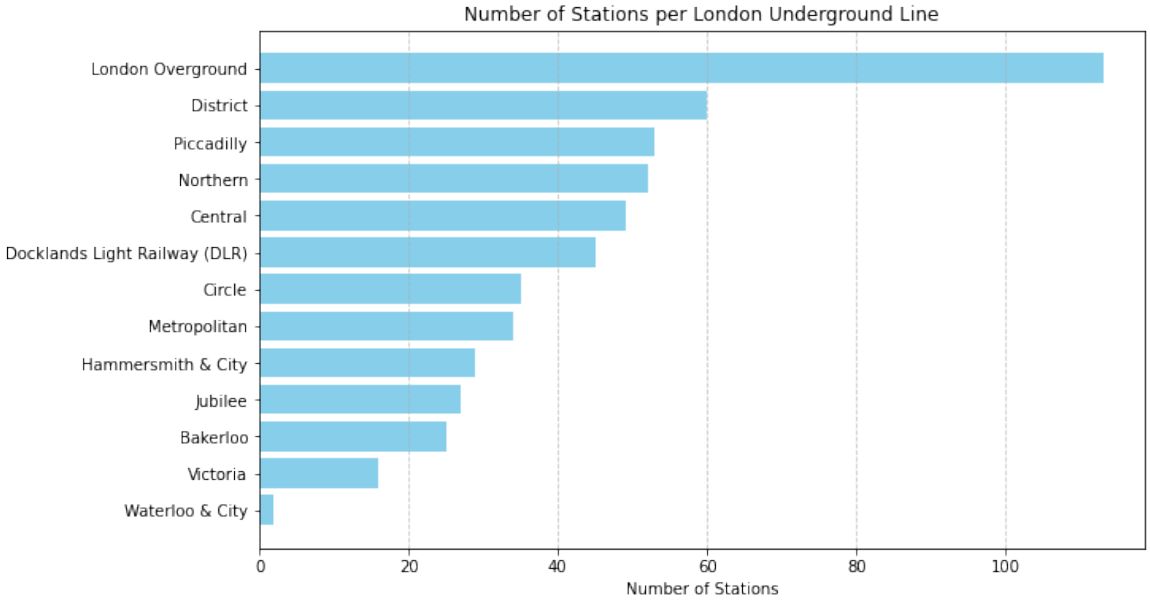
It is identified the following components 11 Tube Lines + other Lines in "London Underground Network":

- **Bakerloo**: 25 stations
- **Central**: 49 stations
- **Circle**: 35 stations
- **District**: 60 stations
- **Hammersmith & City**: 29 stations
- **Jubilee**: 27 stations
- **Metropolitan**: 34 stations
- **Northern**: 52 stations
- **Piccadilly**: 53 stations
- **Victoria**: 16 stations
- **Waterloo & City**: 2 stations
- +Light Rail, Regional Lines etc

The average order of these lines corresponds to 41.62 stations, however there is a high variability between these mean values due to significant outliers, such as the Waterloo & City line, as it can be seen in the bar plot below

In [14]:
```
import matplotlib.pyplot as plt

# Generate Data: London Underground lines and their respective numb
lines = [
    "Waterloo & City", "Victoria", "Jubilee", "Bakerloo", "Hammersm
    "Metropolitan", "Circle", "Northern", "Piccadilly", "Central", "
    "Docklands Light Railway (DLR)", "London Overground"
]
stations = [2, 16, 27, 25, 29, 34, 35, 52, 53, 49, 60, 45, 113]

# Plot
lines_stations = sorted(zip(stations, lines))
stations_sorted, lines_sorted = zip(*lines_stations)
```

```
plt.figure(figsize=(10, 6))
plt.barh(lines_sorted, stations_sorted, color='skyblue')
plt.xlabel('Number of Stations')
plt.title('Number of Stations per London Underground Line')
plt.grid(axis='x', linestyle='--', alpha=0.7)
plt.show()
```

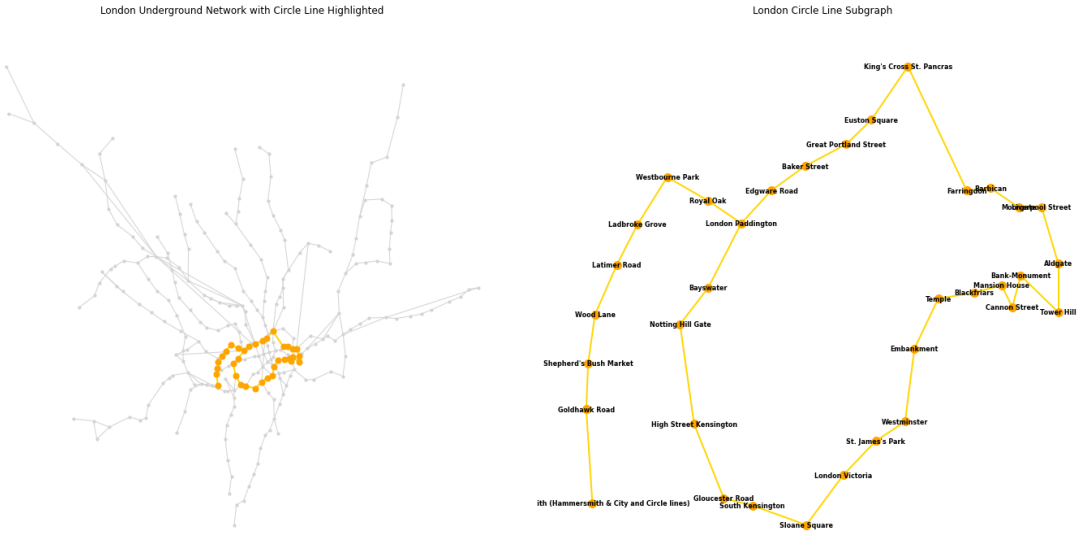


Aditionally, an interesting metro line is Circle Line, as its name refers, it is actually a circle, making the form of a closed walk in a part of the subgraph,an its location goes through the center of the city. In the following plot it is visualy represented.

In [15]:
```
# Define the Circle Line stations
circle_line_stations = [
    "Hammersmith (Hammersmith & City and Circle lines)", "Goldhawk
    "Shepherd's Bush Market", "Wood Lane", "Latimer Road", "Ladbrok
    "Westbourne Park", "Royal Oak", "London Paddington", "Edgware R
    "Baker Street", "Great Portland Street", "Euston Square",
    "King's Cross St. Pancras", "Farringdon", "Barbican", "Moorgate
    "Liverpool Street", "Aldgate", "Tower Hill", "Bank–Monument",
    "Cannon Street", "Mansion House", "Blackfriars", "Temple", "Emb
    "Westminster", "St. James's Park", "London Victoria", "Sloane S
    "South Kensington", "Gloucester Road", "High Street Kensington"
    "Notting Hill Gate", "Bayswater"
]

# Plot Circle Line in Network
circle_line_station_ids = [node for node, name in labels.items() if
circle_line_edges = [(u, v) for u, v in G.edges() if u in circle_li
circle_line_edges = [edge for edge in circle_line_edges if edge not
circle_line_nodes = set(circle_line_station_ids)


fig, axes = plt.subplots(1, 2, figsize=(20, 10))
nx.draw(
    G,
    pos=pos_geo,
```

```
    with_labels=False,
    node_size=10,
    node_color='lightgray',
    edge_color='lightgray',
    ax=axes[0]
)
nx.draw_networkx_edges(
    G,
    pos=pos_geo,
    edgelist=circle_line_edges,
    edge_color='gold',
    width=2,
    ax=axes[0]
)
nx.draw_networkx_nodes(
    G,
    pos=pos_geo,
    nodelist=circle_line_nodes,
    node_color='orange',
    node_size=50,
    ax=axes[0]
)

axes[0].set_title("London Underground Network with Circle Line High


# Plot subgraph
circle_line_subgraph = G.subgraph(circle_line_nodes)

# Plot the Circle Line subgraph
nx.draw(
    circle_line_subgraph,
    pos=pos_geo,
    with_labels=True,
    labels={node: labels[node] for node in circle_line_nodes},
    edgelist=circle_line_edges,
    node_size=100,
    node_color='orange',
    font_size=8,
    font_weight='bold',
    edge_color='gold',
    width=2,
    ax=axes[1]
)
axes[1].set_title("London Circle Line Subgraph")


# Calculate the order (number of nodes) and size (number of edges) 
circle_line_order = len(circle_line_nodes)
circle_line_size = len(circle_line_edges)

print(f"Order of the Circle Line (number of stations): {circle_line

# Adjust layout
plt.tight_layout()
plt.show()
```

```
Order of the Circle Line (number of stations): 35
```



Number of components of the network

As mentioned before, the network is undirected, therefore the number of components in the "London Underground Network" reveals that there is only one connected component, indicating that all stations are part of a single, cohesive network. This suggests that the metro system is designed to ensure complete connectivity, with no isolated or disconnected stations. This is ensuring that every station can be reached from any other station, either directly or via transfer.

The fact that there is a single connected component also reflects the network's design efficiency, and confirms that the network is a weakly connected graph, you can travel between any two vertices by following some set of edges. This means that commuters can travel seamlessly between any two points within the system without encountering unreachable stations. This networks helps in the flexibility of route planning, including multiple tranfer points in more central lines to be able to reach any part of the city that has a metro line.

Additionally, it will enfasises the importance of stations located in the center of the city that can have more traffic, and can serve as critical nodes in maintaining overall network cohesion.

In [16]:
```
#Find connected components
connected_components = list(nx.connected_components(G))
num_components = len(connected_components)

print(f"Number of connected components: {num_components}")
```

```
Number of connected components: 1
```

Diameter

Diameter without weights

The longest shortest path between any two stations in the London Underground network spans 35 vertices (stations). It will indicates, the furthest distance between any two stations, based on the shortest possible route, requires traveling through 35 stations across the city. For the "London Underground Network" those stations (nodes) are going to be "Epping" and "Harrow & Wealdstone"

In [17]:
```python
# Find Diameter of Code
if nx.is_connected(G):
    diameter = nx.diameter(G)
    print(f"The diameter of the connected graph is: {diameter}")
else:
    print("The graph is disconnected.")

# Find the pair of nodes that defines the diameter
if nx.is_connected(G):
    diameter_path = nx.diameter(G)
    longest_shortest_path = nx.periphery(G)

    diameter_nodes = longest_shortest_path[:2]
    print(f"The diameter lies between stations: {diameter_nodes[0]}
```

```
The diameter of the connected graph is: 35
The diameter lies between stations: 345 and 370
```

The following code provides both the sequence of station names along the shortest route and the total distance of travel between the two specified stations, making it useful for analyzing travel routes in the London Undergroung metro system network.

The code calculates the total distance (sum of the weights) of the shortest path using Dijkstra's algorithm, which reflects the total travel distance between the two stations. If no path exists between the nodes or if a node is not found in the graph, appropriate error messages are displayed.

In [18]:
```python
def find_shortest_path(graph, initial_vertex, final_vertex):

    # Find the shortest path
    path = nx.dijkstra_path(graph, source=initial_vertex, target=fi
    path_station_names = [graph.nodes[node].get('label', 'No Label'
    path_length = len(path) - 1
    path_distance = nx.dijkstra_path_length(graph, source=initial_v


    print("Shortest Path (Node IDs):", path)
    print("Shortest Path (Station Names):", path_station_names)
    print("Number of Stations in Path:", path_length)
    print("Total Distance (km):", path_distance)
```

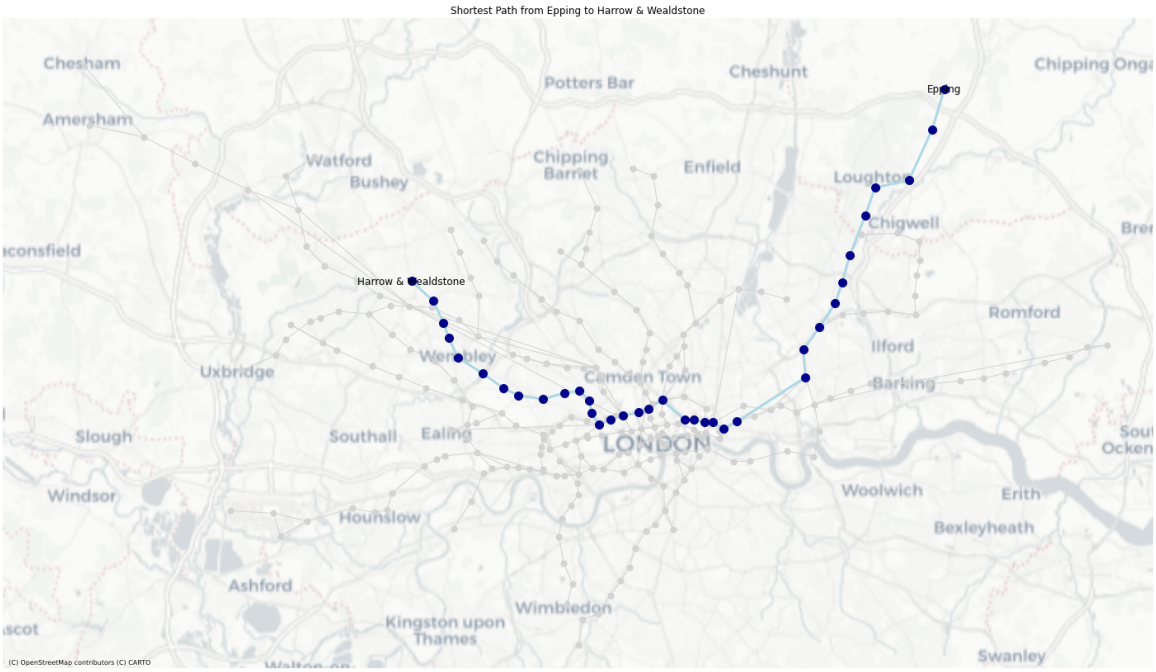
```
find_shortest_path(G, '345', '370')
```

```
Shortest Path (Node IDs): ['345', '494', '329', '411', '302', '537',
'480', '473', '408', '407', '489', '530', '273', '409', '419', '28
3', '349', '400', '347', '360', '280', '341', '441', '514', '412', '
398', '451', '392', '532', '369', '488', '517', '429', '477', '395',
'370']
Shortest Path (Station Names): ['Epping', 'Theydon Bois', 'Debden',
'Loughton', 'Buckhurst Hill', 'Woodford', 'South Woodford', 'Snaresb
rook', 'Leytonstone', 'Leyton', 'Stratford', 'Whitechapel', 'Aldgate
East', 'Liverpool Street', 'Moorgate', 'Barbican', 'Farringdon', "Ki
ng's Cross St. Pancras", 'Euston Square', 'Great Portland Street', '
Baker Street', 'Edgware Road', 'London Paddington', 'Warwick Avenu
e', 'Maida Vale', 'Kilburn Park', "Queen's Park", 'Kensal Green', 'W
illesden Junction', 'Harlesden', 'Stonebridge Park', 'Wembley Centra
l', 'North Wembley', 'South Kenton', 'Kenton', 'Harrow & Wealdston
e']
Number of Stations in Path: 35
Total Distance (km): 64.2756406417444
```

In [19]:
```
def plot_shortest_path(G, start_id, stop_id):

    """
    Plot the shortest path between two nodes in a graph.
    """

    # Calculate shortest path
    try:
        shortest_path = nx.dijkstra_path(G, source=start_id, target=
    except nx.NetworkXNoPath:
        print(f"No path exists between {start_id} and {stop_id}.")
        return

    # Extract edges on shortest path
    path_edges = list(zip(shortest_path[:-1], shortest_path[1:]))

    ### Plot
    fig, ax = plt.subplots(figsize=(25, 25))  # More balanced size

    ax.set_xlim(-0.68, 0.29) # london coords
    ax.set_ylim(51.39, 51.73)

    ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))  # Ad

    # map and network
    ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.Carto
    nx.draw(G, pos=pos_geo, with_labels=False, node_size=50, node_c
            edge_color='lightgray', ax=ax)

    # shortest path
    nx.draw_networkx_edges(G, pos=pos_geo, edgelist=path_edges, edg
    nx.draw_networkx_nodes(G, pos=pos_geo, nodelist=shortest_path,

    # start and stop
    nx.draw_networkx_labels(G, pos=pos_geo, labels={start_id: label

    plt.title(f"Shortest Path from {labels[start_id]} to {labels[st
```

```
    plt.show()




def plot_highlighted_stations(G, pos_geo, station_nodes, labels):
    """
    plot network with highlighted stations
    """

    # Plot the network
    plt.figure(figsize=(12, 8))
    nx.draw(G, pos=pos_geo, with_labels=False, node_size=10, node_c
    nx.draw_networkx_nodes(G, pos=pos_geo, nodelist=station_nodes,

    # Add labels
    nx.draw_networkx_labels(G, pos=pos_geo, labels={node: labels[no

    # cosmetics
    plt.legend(scatterpoints=1, loc='upper left', fontsize=10)
    plt.title("London Tube Network with Highlighted Stations")
    plt.show()
```

In [20]:
```
plot_shortest_path(G, '345', '370')
```



Diameter with weights

In the "London Underground Network", the weighted diameter of the graph can also be calculated. This metric represents the longest shortest path between any two nodes, taking into account the weights assigned to the edges, such as physical distance or travel time.

In this case, using inverse distance as the edge weight (shorter physical distances are given higher weights), the weighted diameter of the graph is

23.68. This is a significant reduction compared to the unweighted diameter, where all edges are treated equally. The use of inverse distance highlights the importance of edge weights in understanding the true structure and efficiency of the network.

The two nodes (stations) that define this weighted diameter are Morden and Hammersmith (specifically the Hammersmith & City and Circle lines). As shown in the map, these stations are located on opposite sides of the city, when the unweighted diameter is calculated, emphasizing the geographic spread and connectivity of the network.

In [21]:
```
# Calculate the shortest paths (weighted)
shortest_path_lengths = dict(nx.all_pairs_dijkstra_path_length(G, w

# Find the diameter
weighted_diameter = max(max(lengths.values()) for lengths in shorte

print(f"The diameter of the graph with weights (inverse distance) i
for source, lengths in shortest_path_lengths.items():
    for target, length in lengths.items():
        if length == weighted_diameter:
            print(f"The two stations corresponding to the weighted
            break
```

```
The diameter of the graph with weights (inverse distance) is: 23.681
40839665368
The two stations corresponding to the weighted diameter are: 421 and
366
```

In [22]:
```
find_shortest_path(G, '421', '366')
```

```
Shortest Path (Node IDs): ['421', '479', '324', '496', '495', '281',
'321', '319', '320', '487', '509', '446', '510', '362', '294', '28
0', '341', '441', '461', '527', '402', '405', '536', '471', '357', '
366']
Shortest Path (Station Names): ['Morden', 'South Wimbledon', 'Collie
rs Wood', 'Tooting Broadway', 'Tooting Bec', 'Balham', 'Clapham Sout
h', 'Clapham Common', 'Clapham North', 'Stockwell', 'Vauxhall', 'Pim
lico', 'London Victoria', 'Green Park', 'Bond Street', 'Baker Stree
t', 'Edgware Road', 'London Paddington', 'Royal Oak', 'Westbourne Pa
rk', 'Ladbroke Grove', 'Latimer Road', 'Wood Lane', "Shepherd's Bush
Market", 'Goldhawk Road', 'Hammersmith (Hammersmith & City and Circl
e lines)']
Number of Stations in Path: 25
Total Distance (km): 29.120318862949997
```

In [23]:
```
plot_shortest_path(G, '421', '366')
```

Adjecency Matrix

The analysis of the "London Underground Network" throught the Adjancency Matrix it is shown a low average 0.009 per node, which is a signal of low connectivity in the network or the metro system. This results goes through the direct behaviour and design of transportation networks, as the statios are mostly connected to few neighbors (in many cases just two , corresponding to the next and previos station) in the same metro line, rather to every other station in the system. This transportation network ephasizes the efficiency through the metro system, therefore having high connectivity can lead to an impractical metro design, as the metro system works as the arms connecting to the center of the city where passengers can create a combination of connections and tranfers to reach their desired station in the city. Therefore, a low connectivity in the "London underground Network" is a characteristic of an optimized network according to the urban mobility.

In [24]:
```
import scipy.io

# Generate the adjacency matrix
adj_matrix = nx.adjacency_matrix(G)
adj_matrix_dense = adj_matrix.todense()

# Calculate the mean
mean_adj_matrix = np.mean(adj_matrix_dense)
print(f"Mean of the adjacency matrix: {mean_adj_matrix}")
```

```
Mean of the adjacency matrix: 0.009022053909556694
```

Network Visualization

With the properties of the "London Underground Network" defined and a

general understanding of the node (stations) and Edge (metro lines) attributes, it is applied some visualization tecniques to gain further understanding of the network

Energy based Visualizations

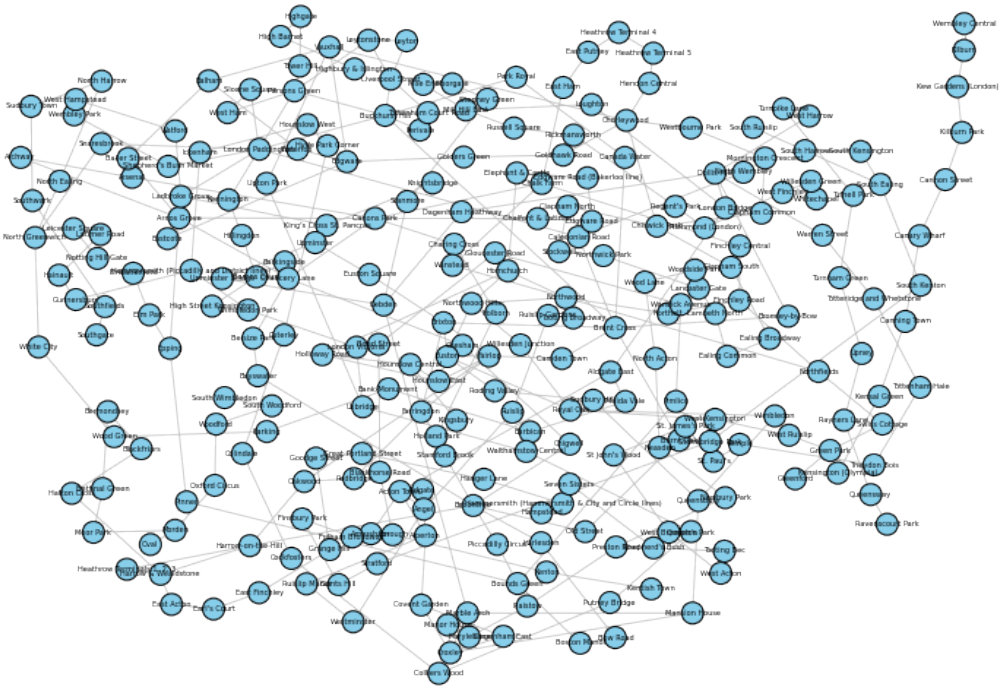
Following graphs layouts are based on energy functions, where the algorithm simulates physical forces to determine node positions. The layout process aims to minimize a global energy function.

The main focus in the "London Underground Network" with the corresponding energy layouts is to reveal clusters, central hubs or important vertex in a clearer visually way.

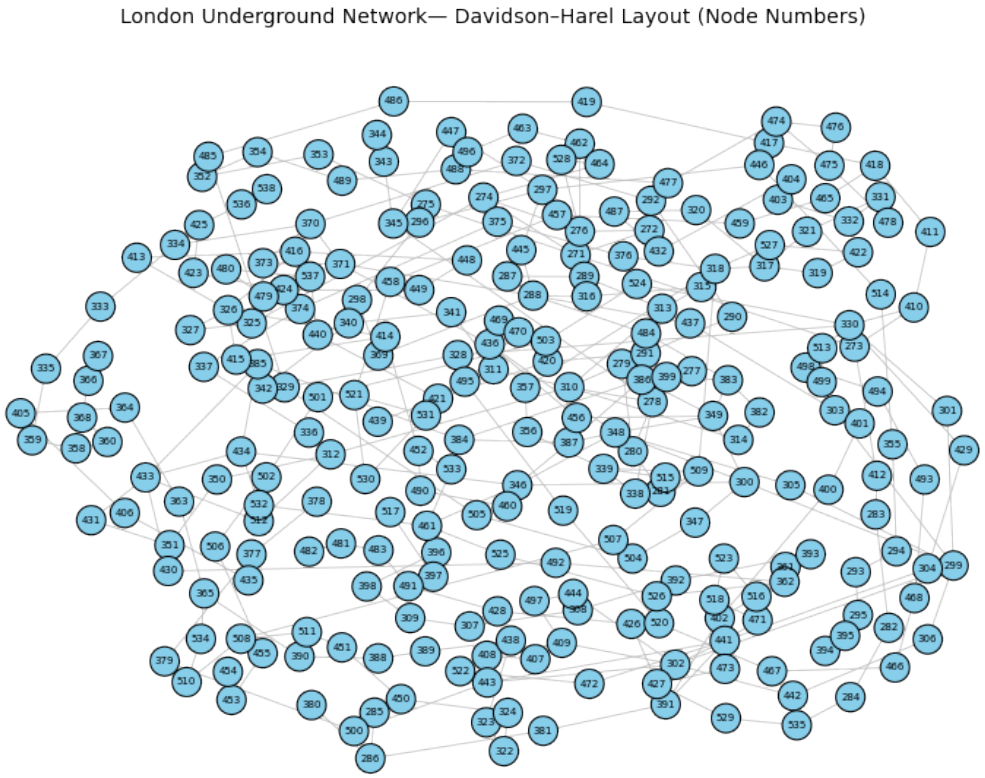
Davidson and Harel layout:

The Davidson and Harel layout will aim for a result with a minimal edge crossing. It is plotted the layouts with the stations names and node number for easier visualization.

In [25]:
```
ig.set_random_number_generator(random.Random(1))

# Prep Graph
edges_i = list(G.edges())
ig_graph = ig.Graph.TupleList(edges_i, directed=False)

if 'label' in list(G.nodes(data=True))[0][1]:
    station_labels = [data['label'] for _, data in G.nodes(data=Tru
else:
    station_labels = [str(n) for n in G.nodes()]

ig_graph.vs["name"] = station_labels

#Compute Davidson–Harel layout
layout_dh = ig_graph.layout("dh")

#Plot the graph
fig, ax = plt.subplots(figsize=(14, 10))
ig.plot(
    ig_graph,
    layout=layout_dh,
    target=ax,
    vertex_label=ig_graph.vs["name"],
    vertex_size=15,
    vertex_color="skyblue",
    edge_color="gray",
    edge_width=0.8,
    vertex_label_size=5,
)
plt.title("London Underground Network – Davidson–Harel Layout", fon
plt.axis('off')
plt.show()
```



In [26]:
```
# Prep
node_ids = list(G.nodes())
ig_graph.vs["name"] = [str(n) for n in node_ids]  # Cast to str for

# Compute Davidson–Harel
layout_dh = ig_graph.layout("dh")

# Plot
fig, ax = plt.subplots(figsize=(14, 10))
ig.plot(
    ig_graph,
    layout=layout_dh,
    target=ax,
    vertex_label=ig_graph.vs["name"],
    vertex_size=20,
    vertex_color="skyblue",
    edge_color="gray",
    edge_width=0.8,
    vertex_label_size=7,
)
plt.title("London Underground Network– Davidson–Harel Layout (Node l
plt.axis('off')
plt.show()
```



The Davidson and Harel layout applied to the "London Underground Network" produces a circular, evenly distributed visualization where no particular emphasis is placed on important stations or clusters. All nodes (stations) are positioned based on force-directed principles, without automatically highlighting central hubs or high-connectivity areas. As a result, key stations with higher degrees are not visually distinguished. To address this, we will manually highlight the stations with the highest degrees to better illustrate their significance within the network.

Let's called again the top 5 nodes with the highest unweighted degrees.

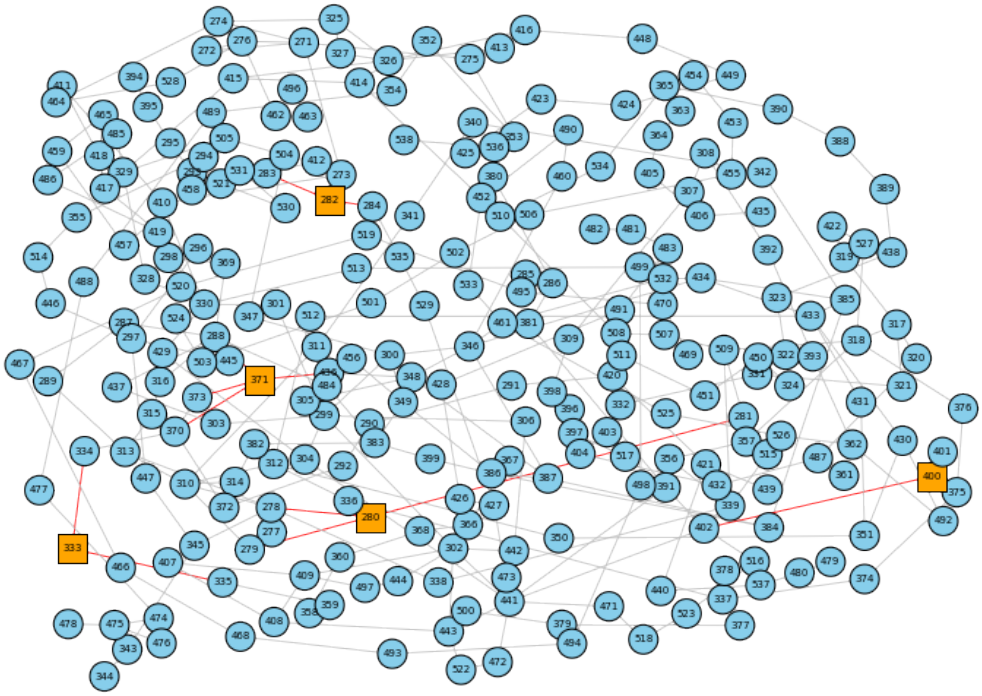
In [27]:
```
#  top 5 nodes with the highest degrees
degrees = dict(G.degree())  # {node: degree}
degree_sequence = [degree for node, degree in degrees.items()]
sorted_nodes_by_degree = sorted(degrees.items(), key=lambda x: x[1]

# top 5 nodes with degrees and labels
print("\nTop 5 nodes with the highest degrees:")
for node, degree in sorted_nodes_by_degree:
    label = G.nodes[node].get('label', 'No Label')
    print(f"Node: {node}, Label: {label}, Degree: {degree}")
```

```
Top 5 nodes with the highest degrees:
Node: 371, Label: Harrow-on-the-Hill, Degree: 8
Node: 280, Label: Baker Street, Degree: 7
Node: 400, Label: King's Cross St. Pancras, Degree: 7
Node: 282, Label: Bank-Monument, Degree: 6
Node: 333, Label: Earl's Court, Degree: 6
```

In [28]:
```
highlight_labels =  ['371','280','400','282','333']
```

```
vertex_colors = ['orange' if name in highlight_labels else 'skyblue
vertex_shapes = ['square' if name in highlight_labels else 'circle'

# Highlight edges connected to top nodes
edge_colors = []
for edge in ig_graph.es:
    src = ig_graph.vs[edge.source]["name"]
    tgt = ig_graph.vs[edge.target]["name"]
    if src in highlight_labels or tgt in highlight_labels:
        edge_colors.append('red')
    else:
        edge_colors.append('gray')

#  Layout and plot
layout_dh = ig_graph.layout("dh")

fig, ax = plt.subplots(figsize=(14, 10))
ig.plot(
    ig_graph,
    layout=layout_dh,
    target=ax,
    vertex_label=ig_graph.vs["name"],
    vertex_size=20,
    vertex_color=vertex_colors,
    vertex_shape=vertex_shapes,
    edge_color=edge_colors,
    edge_width=0.8,
    vertex_label_size=7,
)
plt.title("London Underground Network Davidson–Harel Layout– Top 5 I
plt.axis('off')
plt.show()
```

London Underground Network Davidson-Harel Layout— Top 5 Nodes by Degree Highlighted



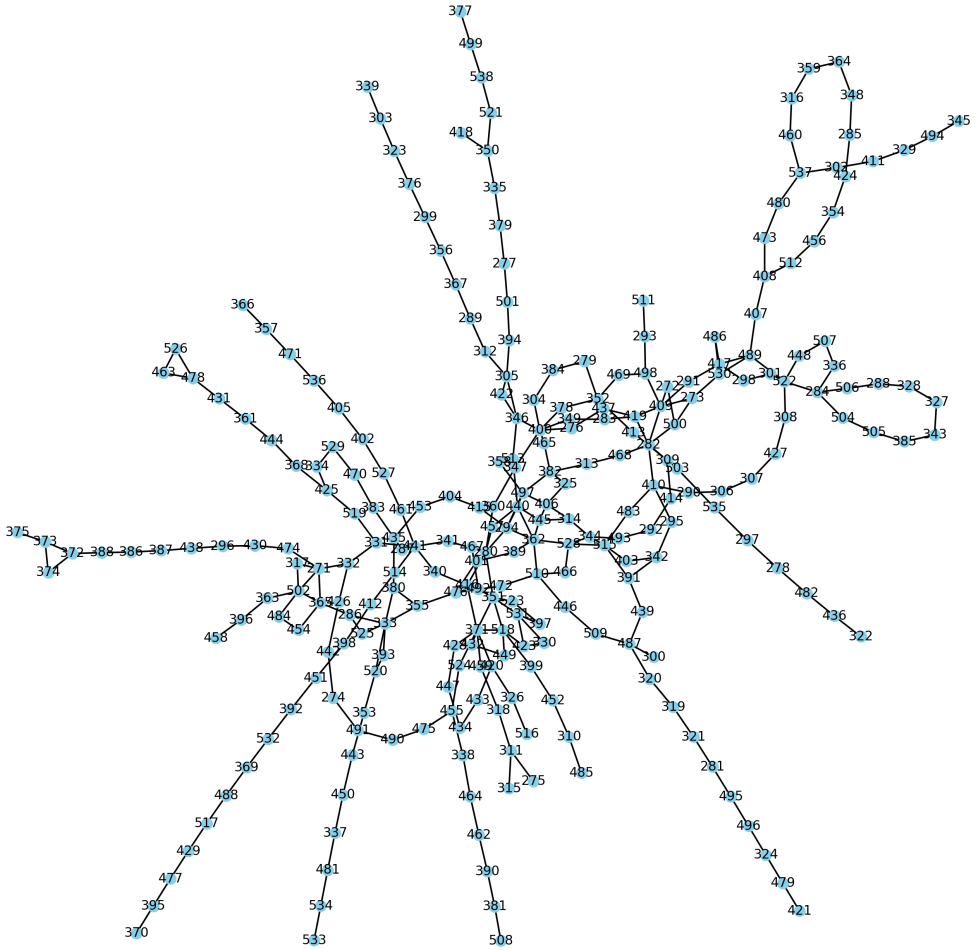
As mentioned, these top 5 stations were not easily identifiable thought visual inspection, therefore the Davidson-Harel layout is not well-suited to get a visual representation of the "London Underground Network". The algorithm layouts in overlapping nodes or distorted relationships, making it difficult to recognize key stations and the overall network structure.

Kamada and Kawai layout:

The Kamada and Kawait layout (another type of energy-based graph visualization method) aims to position nodes so their shortest path distances are preserved as accurately as possible, related nodes are placed closer together and overall structure is more readable.

The Kamada and Kawai layout produces a visualization that closely resembles the actual geographic structure of the London Underground, though not an exact match. By approximating ideal distances between all pairs of stations based on their shortest path lengths, the layout naturally forms a tree-like structure, with several "arms" of the network radiating outward from the central hub stations. This helps to visually emphasize how the main lines converge in central London, reflecting the real-world connectivity and flow of the metro system.

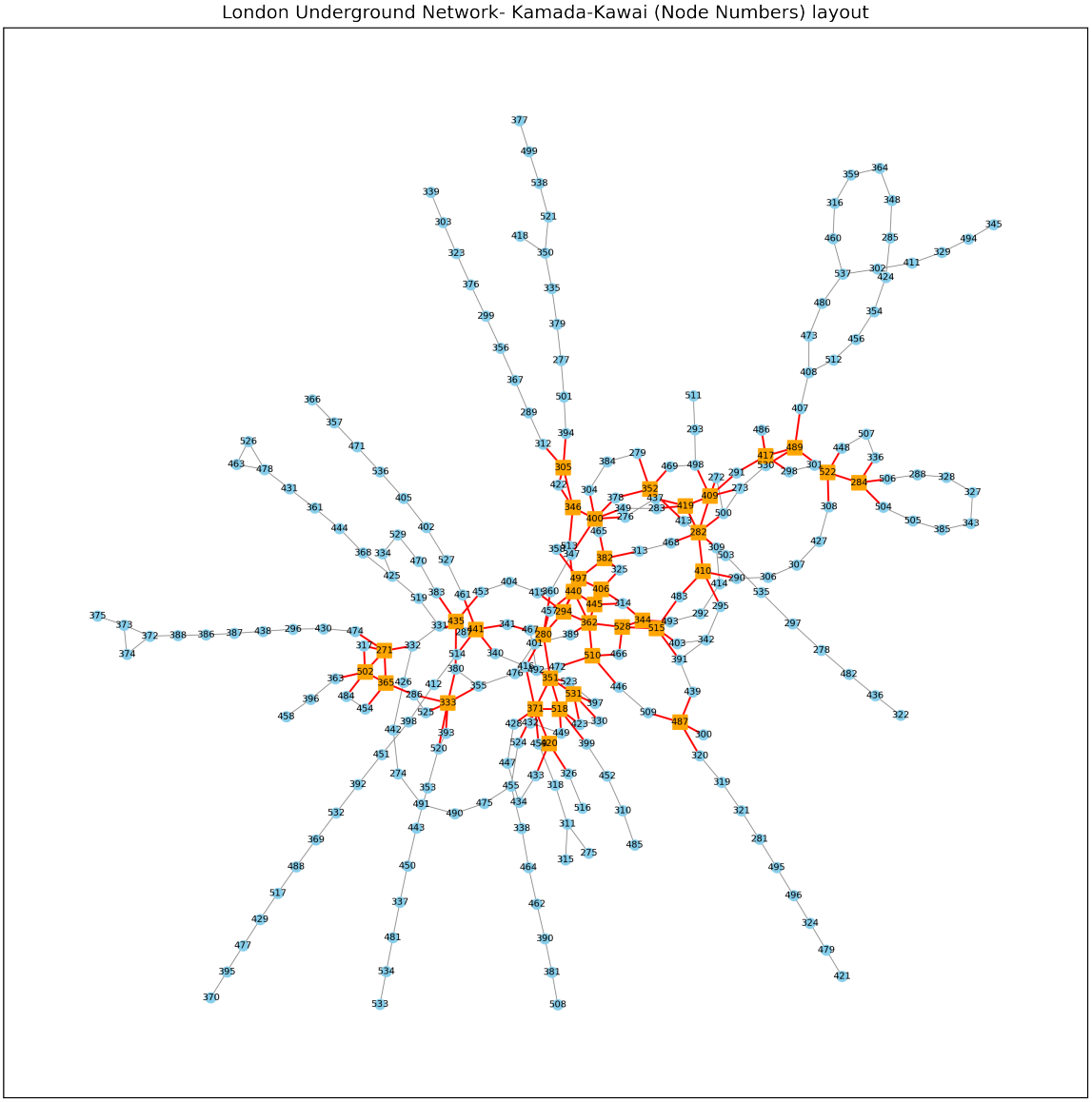
In [29]:
```
# Define layouts
layouts = {
    "London Underground Network - Kamada-Kawai (Node Numbers)": nx.
}

plt.figure(figsize=(30, 20),dpi = 300)
for i, (name, layout_func) in enumerate(layouts.items(), 1):
    pos = layout_func(G)
    plt.subplot(2, 3, i)
    nx.draw(G, pos, with_labels=True, node_size=40, node_color='sky
    plt.title(f"{name} layout")

plt.tight_layout()
plt.show()
```



The following code highlights stations with more than three connections (a degree greater than 3) in the unweighted graph of the "London Underground Network". As seen in the resulting layout, these high-degree stations are predominantly located near the center of the network-city. This distribution aligns with the real-world structure of London, where central stations act as major interchange hubs, facilitating connectivity between multiple lines. These central nodes play a crucial role in ensuring efficient passenger flow across the broader metro system.

In [30]:
```
### Kamda Kawai
# Define layouts
layouts = {
    "London Underground Network- Kamada-Kawai (Node Numbers)": nx.ka
}

# Identify nodes with degree > 3
high_degree_nodes = [n for n in G.nodes if G.degree(n) > 3]
low_degree_nodes = [n for n in G.nodes if G.degree(n) <= 3]

# Identify edges that connect to high-degree nodes
```

```
high_degree_edges = [e for e in G.edges if e[0] in high_degree_node
low_degree_edges = list(set(G.edges) - set(high_degree_edges))

plt.figure(figsize=(30, 20),dpi = 300)
for i, (name, layout_func) in enumerate(layouts.items(), 1):
    pos = layout_func(G)
    plt.subplot(2, 3, i)

    # low-degree nodes
    nx.draw_networkx_nodes(G, pos,
                           nodelist=low_degree_nodes,
                           node_color='skyblue',
                           node_shape='o',
                           node_size=40)

    # high-degree nodes with a different shape and color
    nx.draw_networkx_nodes(G, pos,
                           nodelist=high_degree_nodes,
                           node_color='orange',
                           node_shape='s',  # square
                           node_size=80)

    # edges not connected to high-degree nodes
    nx.draw_networkx_edges(G, pos,
                           edgelist=low_degree_edges,
                           edge_color='gray',
                           width=0.5)

    # edges connected to high-degree nodes
    nx.draw_networkx_edges(G, pos,
                           edgelist=high_degree_edges,
                           edge_color='red',
                           width=1.2)

    nx.draw_networkx_labels(G, pos, font_size=6)
    plt.title(f"{name} layout")

plt.tight_layout()
plt.show()
```



Fruchterman-Reingold

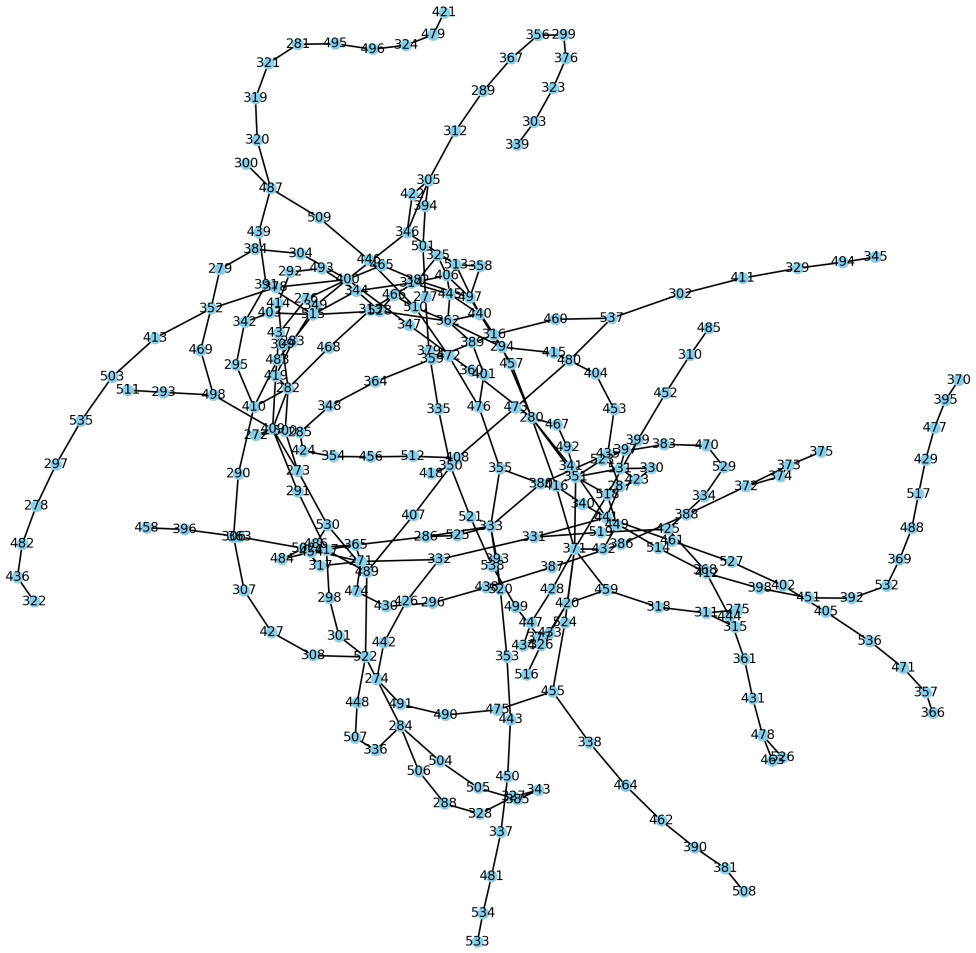
The Fruchterman-Reingol layout, treats nodes like charged particles that repel each other, while edges act like springs that pull connected nodes together, resulting in a layout where related nodes are placed closer together and the overall structure is more visually intuitive.

When applied to the London Underground network, the Fruchterman-Reingold layout reveals a general structure where some "arms" of the network, representing individual metro lines, extend outward with stations connected in a linear fashion. However, the central area appears denser and more tangled, lacking clear clustering or emphasis on specific stations. While the layout does not explicitly highlight important hubs, the visual density at the center suggests the presence of key interchange stations where multiple lines converge. This reflects the real-world role of central London stations as critical connection points within the network.

In [31]:
```
# Define layouts
layouts = {
```

```
    "London Underground Network - Fruchterman-Reingold (Node Number
}

plt.figure(figsize=(30, 20),dpi = 300)
for i, (name, layout_func) in enumerate(layouts.items(), 1):
    pos = layout_func(G, seed = 1)
    plt.subplot(2, 3, i)
    nx.draw(G, pos, with_labels=True, node_size=40, node_color='skyl
    plt.title(f"{name} layout")

plt.tight_layout()
plt.show()
```

London Underground Network - Fruchterman-Reingold (Node Numbers) layout



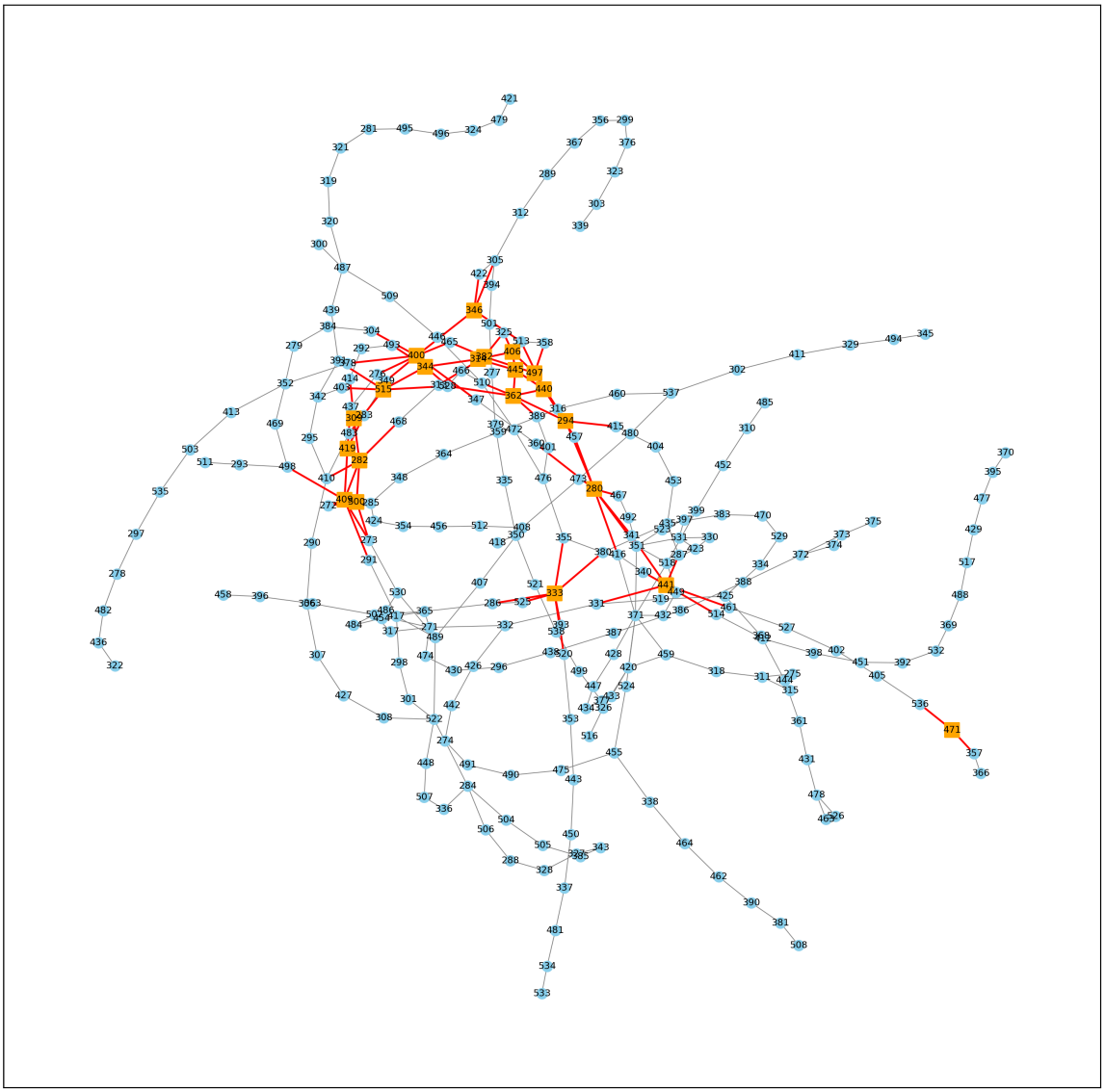
The following code applies the Fruchterman-Reingold force-directed algorithm to visualize the "London Underground Network". It highlights stations (nodes) whose weighted degree calculated as the sum of the inverse distances to neighboring stations exceeds a value of 4. Stations located in central London, where multiple lines intersect and distances between stations are shorter, are more likely to meet this threshold. As expected, several central nodes are visually emphasized in the layout. Notably, station 471, which appears along one of the network's outer branches, is also highlighted. This

prompts a closer investigation into its structure and connections to understand why it stands out in terms of weighted connectivity.

In [32]:
```
# Calculate the weighted degree for each node using inverse of the (
def weighted_degree(graph):
    weighted_degrees = {}
    for node in graph.nodes:
        w_degree = 0
        for neighbor in graph.neighbors(node):
            weight = graph[node][neighbor].get('dist', 1)  # Defaul
            if weight > 0:
                w_degree += 1 / weight
        weighted_degrees[node] = w_degree
    return weighted_degrees

# Classify nodes based on weighted degree
weighted_degrees = weighted_degree(G)
threshold = 4
high_degree_nodes = [n for n, d in weighted_degrees.items() if d > 
low_degree_nodes = [n for n in G.nodes if n not in high_degree_node

# Classify edges connected to high-degree nodes
high_degree_edges = [e for e in G.edges if e[0] in high_degree_node
low_degree_edges = list(set(G.edges) - set(high_degree_edges))

# Plot
layouts = {
    "London Tube Network - Fruchterman-Reingold (Weighted Degree > 
}

plt.figure(figsize=(30, 20),dpi = 300)
for i, (name, layout_func) in enumerate(layouts.items(), 1):
    pos = layout_func(G, seed = 1)
    plt.subplot(2, 3, i)

    # low-degree nodes
    nx.draw_networkx_nodes(G, pos,
                           nodelist=low_degree_nodes,
                           node_color='skyblue',
                           node_size=40,
                           node_shape='o')

    # high-degree nodes
    nx.draw_networkx_nodes(G, pos,
                           nodelist=high_degree_nodes,
                           node_color='orange',
                           node_size=80,
                           node_shape='s')

    # edges not connected to high-degree nodes
    nx.draw_networkx_edges(G, pos,
                           edgelist=low_degree_edges,
                           edge_color='gray',
                           width=0.5)
```

```
    # edges connected to high-degree nodes
    nx.draw_networkx_edges(G, pos,
                           edgelist=high_degree_edges,
                           edge_color='red',
                           width=1.2)

    # labels
    nx.draw_networkx_labels(G, pos, font_size=6)

    plt.title(f"{name}")

plt.tight_layout()
plt.show()
```

London Tube Network - Fruchterman-Reingold (Weighted Degree > 4)



Upon further analysis of node 471, which corresponds to Shepherd's Bush Market station, we observe that its degree is 2. This means the station has 2 edges connecting it to other stations in the network. However, the weighted degree of the station is significantly higher, at 4.315. The weighted degree is calculated based on the inverse of the distance to neighboring stations, which is used to measure the strength of its connections. Specifically, the inverse of the distance (smaller distances contribute more heavily) reflects how "close"

a node is to its neighbors. Lower distances between stations lead to higher weighted degrees, meaning that Shepherd's Bush Market is relatively close to its connected stations compared to other nodes with higher distances, making it a more strongly connected node in terms of proximity.

This closer relationship is further emphasized in the plot of "London Tube Network - Fruchterman-Reingold (Weighted Degree > 4)". In this plot, stations with weighted degrees greater than 4 are highlighted, despite Shepherd's Bush Market station having only 2 direct connections (degree without weights), its strong weighted degree indicates that those connections are particularly close, contributing to its prominence in the network.

In [33]:
```
# Get degree
node = '471'
degree_ = G.degree[node]
print(f"Degree of node {node}:", degree_)

label_ = G.nodes[node].get('label', 'No label found')
print(f"Label of node {node}:", label_)

# Retrieve weighted degree of node 471
weighted_degree_ = weighted_degrees.get(node, "Node not found")
print(f"Weighted degree of node {node}:", weighted_degree_)
```

```
Degree of node 471: 2
Label of node 471: Shepherd's Bush Market
Weighted degree of node 471: 4.315239636255769
```

When using energy-based layouts, it becomes evident that the Kamada-Kawai algorithm provides a visualization that goes closely to the real spatial structure of the "London Underground Network", especially when plotted using node coordinates. However, overall, energy-based visualizations do not offer a better understanding of the network than the simple geographical layout initially used. Since the network is already well-defined and each station has accurate geographic coordinates, these abstract layouts add little new insight into its structure. But, they can be useful for offering a quick visual indication of node centrality, depending on the specific energy function applied. It may be valuable to revisit these plots after calculating centrality measures, as the layouts might better highlight structural roles and relationships within the network under that context.

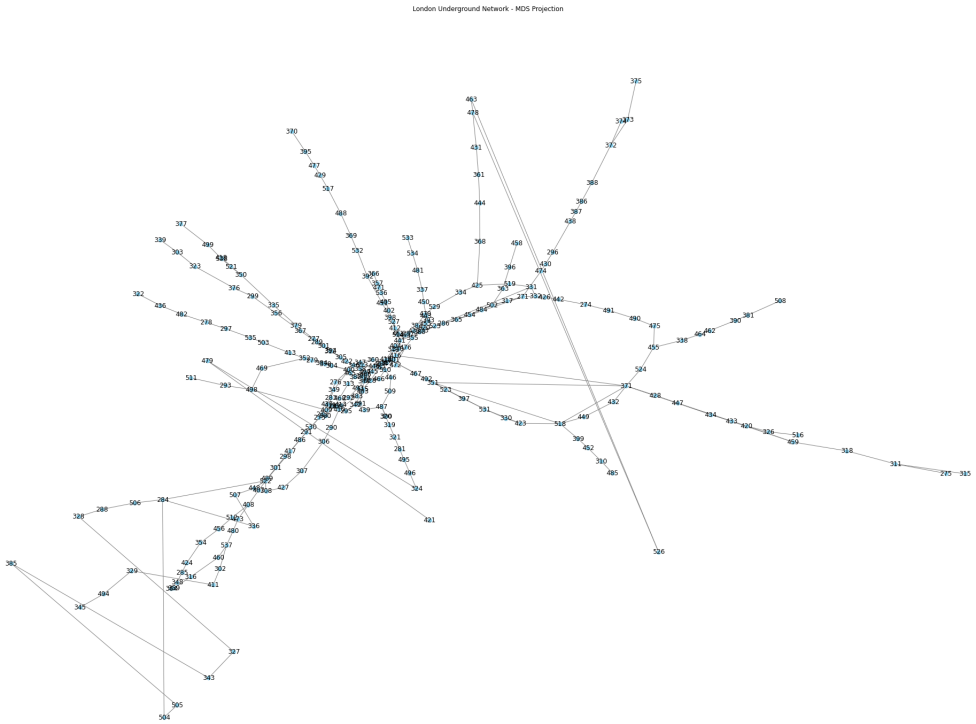
Multi Dimensional Scaling (MDS)

Multi Dimensional Scaling is a distance matrix based dimension reduction technique. To apply it to the "London Underground Network", first it is generate the distance matrix by solving shortest path issues for each pair of nodes.

In [34]:
```
# Generate distance Matrix (Compute the shortest path distances betu
distance_matrix = dict(nx.all_pairs_dijkstra_path_length(G, weight=

# Convert to df
distance_df = pd.DataFrame.from_dict(distance_matrix, orient='index
distance_df = distance_df.reindex(index=G.nodes(), columns=G.nodes(
```

With the distance matrix set up, the MDS methods is imported and then is applied them to distance matrix. The resulting visualization has a similar tree like structure, with most nodes close distances in the network also remain close in the MDS Visualization. However, it is also identified some clearly distorted node, like node 526, with no clear explanation on why they appear to be so distorted. In general the first MDS Coordinate appears to have been in some ways the longitude but flipped (e.g. stations previously to the right now more to the left and vice versa), while the second MDS coordinate is not very insightful, with some stations previously in the North remaining, while others are moved to the "south".

In [35]:
```
from sklearn.manifold import MDS

# Run MDS
distance_array = distance_df.values
mds = MDS(n_components=2, dissimilarity='precomputed', random_state=
mds_coordinates = mds.fit_transform(distance_array)
mds_df = pd.DataFrame(mds_coordinates, columns=['MDS1', 'MDS2'], in


# Plot MDS (first two components)
plt.figure(figsize=(30, 20))
mds_positions = {str(node): (row['MDS1'], row['MDS2']) for node, ro
nx.draw(
    G,
    pos=mds_positions,
    with_labels=True,
    node_size=50,
    node_color='skyblue',
    edge_color='gray'
)
plt.title("London Underground Network - MDS Projection")
plt.show()
```



In some way, the result can be seen as an attempt to reconstruct the attributes longitude and latitude via first and second MDS coordinate from the nodes pairwise distances. As however the precise geographical positions of the nodes are known already through their longitude and latitude, MDS appears not to be able to add significant insights.

Centrality Measures

The following section contains the analysis into the centrality of nodes and edges.

Degree distribution

Degrees without weights

The average degree is calculated to be 2.42, which aligns with expectations. As most stations in the metro network are typically connected by a single line, with each station having two adjacent connections: one to the previous station and one to the next. Such a structure is common in metro systems, where the majority of stations are part of a linear sequence of stops

In [36]:
```
# Calculate the average degree of the graph
average_degree = sum(dict(G.degree()).values()) / len(G.nodes())

print(f"Average degree of the network: {average_degree:.2f}")
```

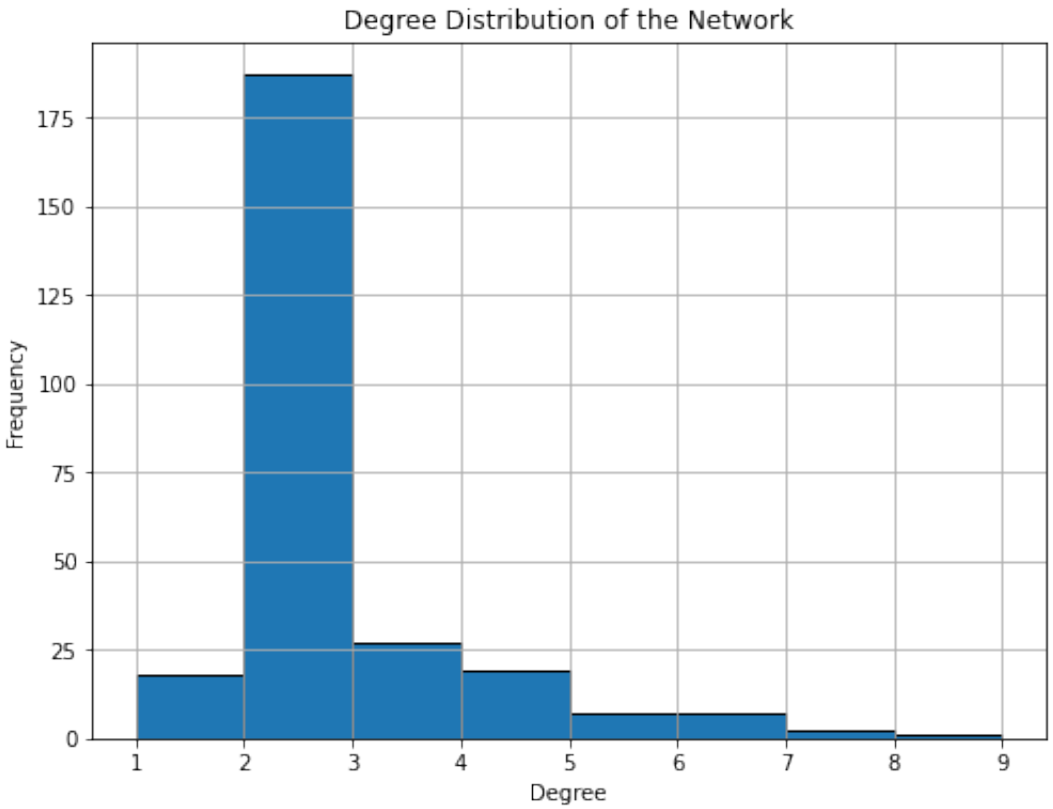
```
Average degree of the network: 2.42
```

In general, we observe that that stations have a degree of two, being a pass

through station on one of the lines. Then, there is a few nodes having degree 1 corresponding to end stops of lines and other stations with degree 3+ connecting two or more lines.

The top 5 nodes with the highest degrees in the network are Harrow-on-the-Hill, Baker Street, Bank-Monument, King's Cross St. Pancras, and Earl's court. Harrow-on-the-Hill stands out with the highest degree of 8, indicating it has the most connections in the network. The other two stations—Baker Street and King's Cross St. Pancras have a degree of 7, showing they are all well-connected with a similar number of edges. Following the other two reamingn stations Bank-monument and Earl's Court have a degree of 6, in general, these stations are central hubs in the network, with varying but significant levels of connectivity. These stations are all above the average degree of the network, which is 2.42. Their higher degrees suggest they are significantly more connected than the average station, reflecting their importance in facilitating movement and providing multiple transport links. Some initial assumptions about their locations and roles in London transportation help provide context, a more detailed analysis would involve evaluating commuter traffic, station facilities, and transport links to further understand their importance within the network.

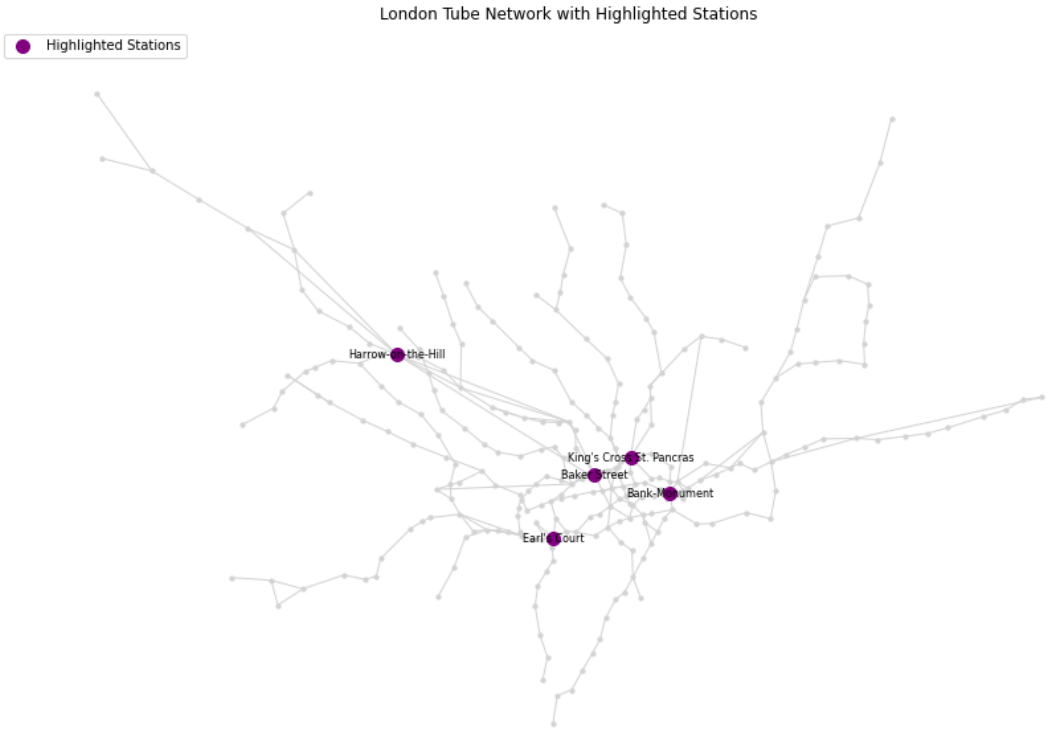
In [37]:
```
# Degree distribution (histogram)
degrees = dict(G.degree())

degree_sequence = [degree for node, degree in degrees.items()]

plt.figure(figsize=(8, 6))
plt.hist(degree_sequence, bins=range(min(degree_sequence), max(degr
plt.title('Degree Distribution of the Network')
plt.xlabel('Degree')
plt.ylabel('Frequency')
plt.grid(True)
plt.show()

# top 5 nodes with highest degrees
sorted_nodes_by_degree = sorted(degrees.items(), key=lambda x: x[1]

# Print top 5 nodes with their degrees and labels
print("\nTop 5 nodes with the highest degrees:")
for node, degree in sorted_nodes_by_degree:
    label = G.nodes[node].get('label', 'No Label')
    print(f"Node: {node}, Label: {label}, Degree: {degree}")
```




```
Top 5 nodes with the highest degrees:
Node: 371, Label: Harrow-on-the-Hill, Degree: 8
Node: 280, Label: Baker Street, Degree: 7
Node: 400, Label: King's Cross St. Pancras, Degree: 7
Node: 282, Label: Bank-Monument, Degree: 6
Node: 333, Label: Earl's Court, Degree: 6
```

Overall, these stations indeed are more central to the network in a connectivity sense, as they represent the hubs at which passengers of the metro system can switch between lines and will thus see a substantial percentage of all passangers moving through those hubs.

In [38]:
```
# top 5 nodes by degree
top_5_degree_nodes = [node for node, degree in sorted_nodes_by_degr
plot_highlighted_stations(G, pos_geo, top_5_degree_nodes, labels)
```



Degrees with weights

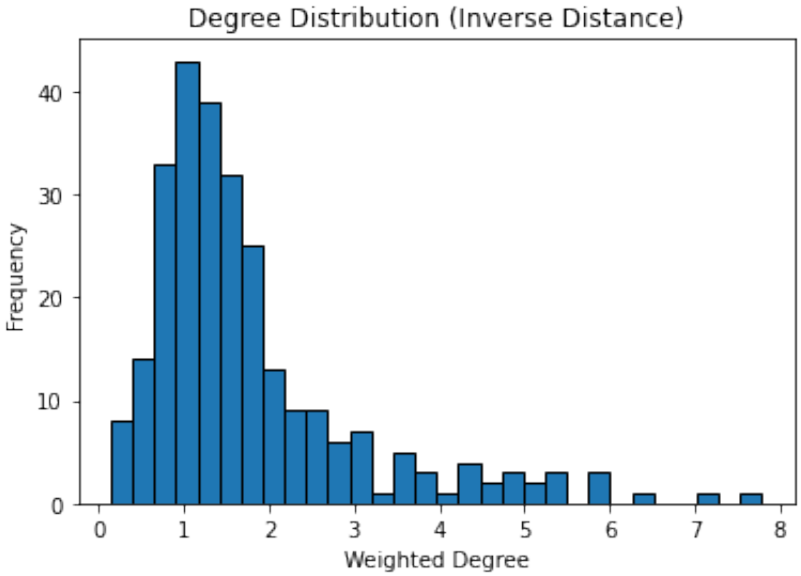
The network's degree distribution reveals an average weighted degree of 1.77, indicating that, on average, each station has a relatively modest number of strong connections when considering the inverse of the distance as the weight. The arcs weighted by their inverse distance can in this case be interpreted as the geographical closeness of the station to other stations in the network, as the inverse of a small distance yields a large weight. Then a large sum of these weights shows that metro stations have either a few stations that are very close to it or many stations connected to it but that might be a bit further away.

Notably, the top 5 stations with the highest weighted degrees, which are Bank-Monument (7.79), Leicester Square (7.18), Embankment (6.36), Charing Cross (5.92), and Waterloo (5.77), are all located in central and highly trafficked areas of the metro system. These stations exhibit significantly higher connectivity strength, suggesting they serve as major hubs where multiple metro lines converge. Their elevated weighted degrees reflect their crucial roles in the network, as they facilitate efficient transfers and high passenger flow. In the metro system is expected that the central stations are designed to handle high volumes of passengers and connections across various lines.

In [39]:
```
# Weighted degree
def weighted_degree(graph, labels):
    weighted_degrees = {}
    for node in graph.nodes:
```

```
        weighted_degree = 0
        for neighbor in graph.neighbors(node):
            #(inverse of distance as weight)
            weight = graph[node][neighbor].get('dist', 1)
            weighted_degree += 1 / weight
        weighted_degrees[node] = weighted_degree
    return weighted_degrees

weighted_degrees = weighted_degree(G, labels)

# average degree
average_degree = sum(weighted_degrees.values()) / len(weighted_degr
print(f"Average weighted degree: {average_degree:.2f}")

# top 5 nodes
top_5_stations = sorted(weighted_degrees.items(), key=lambda x: x[1
print("Top 5 stations with highest weighted degrees:")
for node, degree in top_5_stations:
    station_name = labels.get(node, "Unknown Station")  # Get the s
    print(f"Station: {station_name} (ID: {node}), Weighted Degree:


# Plot
all_degrees = list(weighted_degrees.values())

plt.hist(all_degrees, bins=30, edgecolor='black')
plt.title("Degree Distribution (Inverse Distance)")
plt.xlabel("Weighted Degree")
plt.ylabel("Frequency")
plt.show()
```

```
Average weighted degree: 1.77
Top 5 stations with highest weighted degrees:
Station: Bank–Monument (ID: 282), Weighted Degree: 7.79
Station: Leicester Square (ID: 406), Weighted Degree: 7.18
Station: Embankment (ID: 344), Weighted Degree: 6.36
Station: Charing Cross (ID: 314), Weighted Degree: 5.92
Station: Waterloo (ID: 515), Weighted Degree: 5.77
```



In [40]:
```
print("Top 5 nodes with the highest weighted degrees:")
```

```
for node, degree in sorted(weighted_degrees.items(), key=lambda x: x
    print(f"Node: {node} ({labels[node]}), Weighted Degree: {degree
```

```
Top 5 nodes with the highest weighted degrees:
Node: 282 (Bank–Monument), Weighted Degree: 7.79
Node: 406 (Leicester Square), Weighted Degree: 7.18
Node: 344 (Embankment), Weighted Degree: 6.36
Node: 314 (Charing Cross), Weighted Degree: 5.92
Node: 515 (Waterloo), Weighted Degree: 5.77
```

Meanwhile the top stations with a weighted degree have less significance as the closeness of a station to other stations does not necessarily make it more central, even though the spacing of stations in the center becomes less dense, yielding higher weights for stations in the geographical center of the city.

In [41]:
```
#Get top 5 nodes by degree weighted
top_5_stations_g = sorted(weighted_degrees.items(), key=lambda x: x
top_5_degree_nodes = [node for node, degree in top_5_stations_g]

plot_highlighted_stations(G, pos_geo, top_5_degree_nodes, labels)
```

Closeness Centrality

The closeness centrality, being 1 over the sum of (shortest) distances of the one node to every other node in the system, gives a metric that defines centrality in the sense of closeness to all other nodes in the system. The top stations with that metric are very close to each other and in a sense in the "center of gravity" of the network, roughly corresponding to the city center. In the context of public transport, these however not necessarily coincide with

the most important stations as in this case, the top 5 stations yielded do not correspond with the hub stations that the degree analysis enfasizes.

In [42]:
```
# Calculate closeness centrality for all nodes and sort
closeness_centrality = nx.closeness_centrality(G, distance='dist')
top_closeness_stations = sorted(closeness_centrality.items(), key=la

# Print top 5
print("Top 5 stations with highest closeness centrality:")
for node, centrality in top_closeness_stations:
    station_name = labels.get(node, "Unknown Station")
    print(f"Node: {node} ({labels[node]}), Closeness Centrality: {c

top_closeness_ids = [station[0] for station in top_closeness_statio
plot_highlighted_stations(G, pos_geo, top_closeness_ids, labels)
```

```
Top 5 stations with highest closeness centrality:
Node: 440 (Oxford Circus), Closeness Centrality: 0.0637
Node: 362 (Green Park), Closeness Centrality: 0.0633
Node: 294 (Bond Street), Closeness Centrality: 0.0630
Node: 280 (Baker Street), Closeness Centrality: 0.0626
Node: 445 (Piccadilly Circus), Closeness Centrality: 0.0626
```

Betweenness Centrality: Nodes

With hubs as the most important stations in the network, betweenness centrality as the percentage of shortest paths from all pairs of nodes going through a specifif node is interesting as hubs are expected to have a high value for this measure by connecting different metro lines. As expected, the top 5 stations are similiar to those from the top 5 stations by degree and represent known hub stations where multiple lines connect.

In [43]:
```
# Calculate betweenness centrality for all nodes and sort
betweenness_centrality = nx.betweenness_centrality(G, weight='dist'
top_betweenness_stations = sorted(betweenness_centrality.items(), k

# Print top 5
print("Top 5 stations with highest betweenness centrality:")
for node, centrality in top_betweenness_stations:
    station_name = labels.get(node, "Unknown Station")
    print(f"Node: {node} ({labels[node]}), Betweenness Centrality:

plot_highlighted_stations(G, pos_geo, [station[0] for station in to
```

```
Top 5 stations with highest betweenness centrality:
Node: 280 (Baker Street), Betweenness Centrality: 0.3052
Node: 409 (Liverpool Street), Betweenness Centrality: 0.2675
Node: 400 (King's Cross St. Pancras), Betweenness Centrality: 0.2555
Node: 333 (Earl's Court), Betweenness Centrality: 0.2218
Node: 441 (London Paddington), Betweenness Centrality: 0.2008
```

Eigenvector Centrality

Eigenvector centrality as a metric drawn from the Eigenvector-decomposition of the adjecency matrix with the idea of quantifying the importance a node by taking to account the importance of neighboring nodes (For Social Networks: Having few very influencial friends can be as good as having many friends) yields dificult to interprete results for the metro network. The resulting top 5 Stations are not significant as stations in the network and appear to have the main characteristic of being geographically central with 3 stations identical to the stations yielded from closeness centrality. This metric, designed for social network analysis appears to not be fit to make meaningful interpretations to the network at hand.

In [44]:
```
# Calculate eigenvector centrality and get top 5 nodes
eigenvector_centrality = nx.eigenvector_centrality(G, max_iter=1000
top_eigenvector_stations = sorted(eigenvector_centrality.items(), k

print("Top 5 stations with highest eigenvector centrality:")
for node, centrality in top_eigenvector_stations:
    station_name = labels.get(node, "Unknown Station")
    print(f"Node: {node} ({station_name}), Eigenvector Centrality:

plot_highlighted_stations(
    G,
    pos_geo,
    [node for node, _ in top_eigenvector_stations],
    labels
)
```

```
Top 5 stations with highest eigenvector centrality:
Node: 440 (Oxford Circus), Eigenvector Centrality: 0.3744
Node: 294 (Bond Street), Eigenvector Centrality: 0.3369
Node: 362 (Green Park), Eigenvector Centrality: 0.3289
Node: 497 (Tottenham Court Road), Eigenvector Centrality: 0.2764
Node: 280 (Baker Street), Eigenvector Centrality: 0.2738
```

Pagerank

Pagerank, being similar to the degree in this case of an undirected graph similar to degree (in-degree), for undirected graphs similar to the degree, correspondingly yields 3 identical stations to the degree analysis although with a less well interpretable metric. Overall the metrics appear to be low due to low overall connectivity in the network as shown in the adjecency matrix.

In [45]:
```
# Calculate PageRank for all nodes
pagerank = nx.pagerank(G)
top_pagerank_stations = sorted(pagerank.items(), key=lambda x: x[1]

print("Top 5 stations with highest PageRank:")
for node, rank in top_pagerank_stations:
    station_name = labels.get(node, "Unknown Station")
    print(f"Node: {node} ({station_name}), PageRank: {rank:.4f}")

# Plot
top_pagerank_ids = [station[0] for station in top_pagerank_stations
plot_highlighted_stations(G, pos_geo, top_pagerank_ids, labels)
```

```
Top 5 stations with highest PageRank:
Node: 371 (Harrow-on-the-Hill), PageRank: 0.0097
Node: 400 (King's Cross St. Pancras), PageRank: 0.0088
Node: 280 (Baker Street), PageRank: 0.0079
Node: 441 (London Paddington), PageRank: 0.0079
Node: 333 (Earl's Court), PageRank: 0.0073
```

Betweenness Centrality: Edges

As done previously for nodes, for Edges the betweenness centrality can be evaluated finding the most used edges when going from all nodes to every other nodes via the shortest path. While this might give insights into the most used network sections for a evenly used transportation network, overall the meaning of this network is not as significant as for the nodes as the traffic on specific edges will depend more on the points of interest in the city of london than on the theoretical shortest paths to all other points. None the less, unsurprisinly, many of the top 5 Edges start/end in a previously as central identified node.

In [46]:
```
# Calculate top 5 Edges via Betweenness centrality
edge_betweenness_centrality = nx.edge_betweenness_centrality(G)
top_betweenness_edges = sorted(edge_betweenness_centrality.items(),

print("Top 5 edges with highest betweenness centrality:")
for edge, centrality in top_betweenness_edges:
    station_1 = labels.get(edge[0], "Unknown Station")
    station_2 = labels.get(edge[1], "Unknown Station")
    print(f"Edge: {edge} (Connecting {station_1} and {station_2}), B
```

```
Top 5 edges with highest betweenness centrality:
Edge: ('280', '341') (Connecting Baker Street and Edgware Road), Bet
weenness Centrality: 0.2545
Edge: ('341', '441') (Connecting Edgware Road and London Paddingto
n), Betweenness Centrality: 0.2515
Edge: ('280', '294') (Connecting Baker Street and Bond Street), Betw
eenness Centrality: 0.2094
Edge: ('331', '441') (Connecting Ealing Broadway and London Paddingt
on), Betweenness Centrality: 0.1911
Edge: ('280', '360') (Connecting Baker Street and Great Portland Str
eet), Betweenness Centrality: 0.1722
```

Network Cohesion

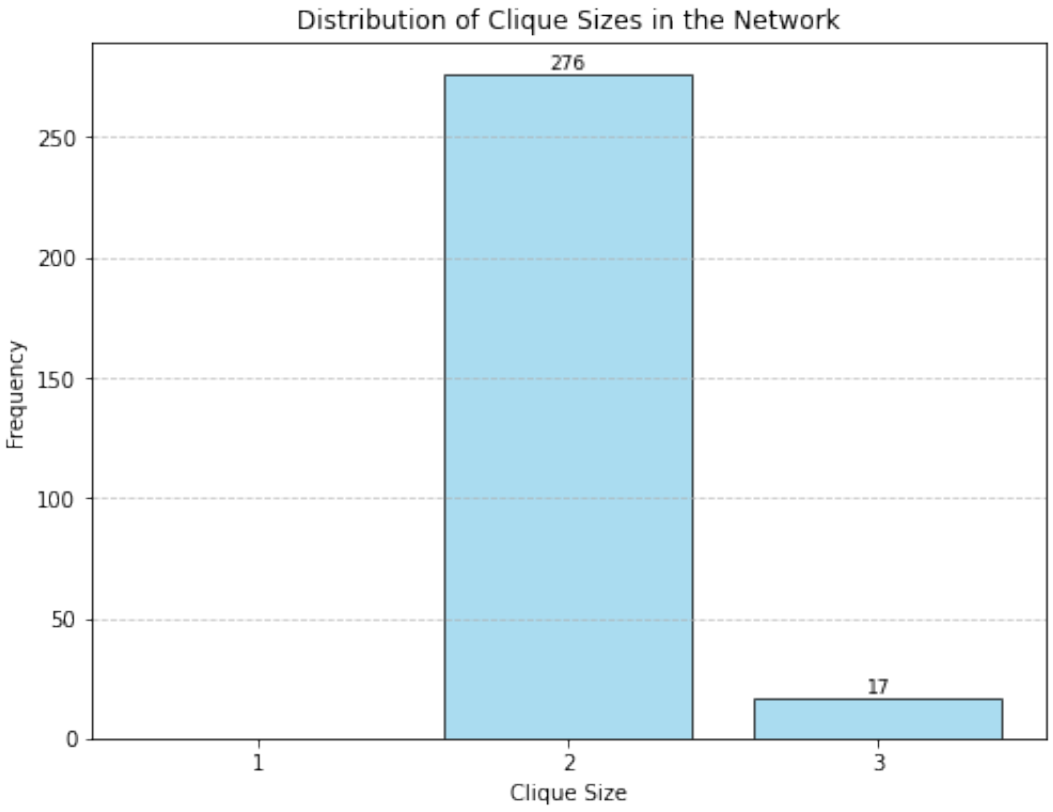
Local Density

Distribution of Clique Sizes

The analysis of clique sizes in the "London Underground Network" it is performed to reveal important structural characteristics of the system. The network contains 276 cliques of size 2, which correspond to direct connections between pairs of stations, as mentioned before, one station will mostly have just 2 neighbours, the next and the previous station, therefore this number is expected, most of the stations with 2 cliques. These pairwise connections form the basic structure of the "London Underground Network", reflecting its efficient, metro line and arms design where stations serve as sequential stops along each line.

In the other hand, only 17 cliques of size 3 were identified, indicating that fully interconnected groups of three stations, where each station is directly connected to the other two are rare. These triangular structures are more likely to occur in densely connected hubs or transfer points, such as central London or areas like Heathrow Airport, where multiple lines intersect. This low presence of cliques of size 3 reinforces the idea that the network as being a transportation network, is not highly meshed, but rather a tree design with arms that encounters in the center of the city as mentioned and shown in the previous part of the analysis. This pattern also contributes toa simplicity and efficiency system for the metro in the city, where passengers can reach any point of the network where transfers are possible in central stations or nodes.

In [47]:
```
# Get cliques
cliques = list(nx.find_cliques(G))
clique_sizes = [len(clique) for clique in cliques]

# plot dist
plt.figure(figsize=(8, 6))
bars = plt.bar(range(1, max(clique_sizes) + 1),
               [clique_sizes.count(i) for i in range(1, max(clique_
               width=0.8,
               edgecolor='black',
               color='skyblue',
               alpha=0.7)

# cosmetics
plt.title('Distribution of Clique Sizes in the Network')
plt.xlabel('Clique Size')
plt.ylabel('Frequency')
plt.xticks(range(1, max(clique_sizes) + 1))
plt.grid(axis='y', linestyle='--', alpha=0.7)

for bar in bars:
    height = bar.get_height()
    if height > 0:
        plt.text(bar.get_x() + bar.get_width() / 2, height + 0.5, s
plt.show()
```



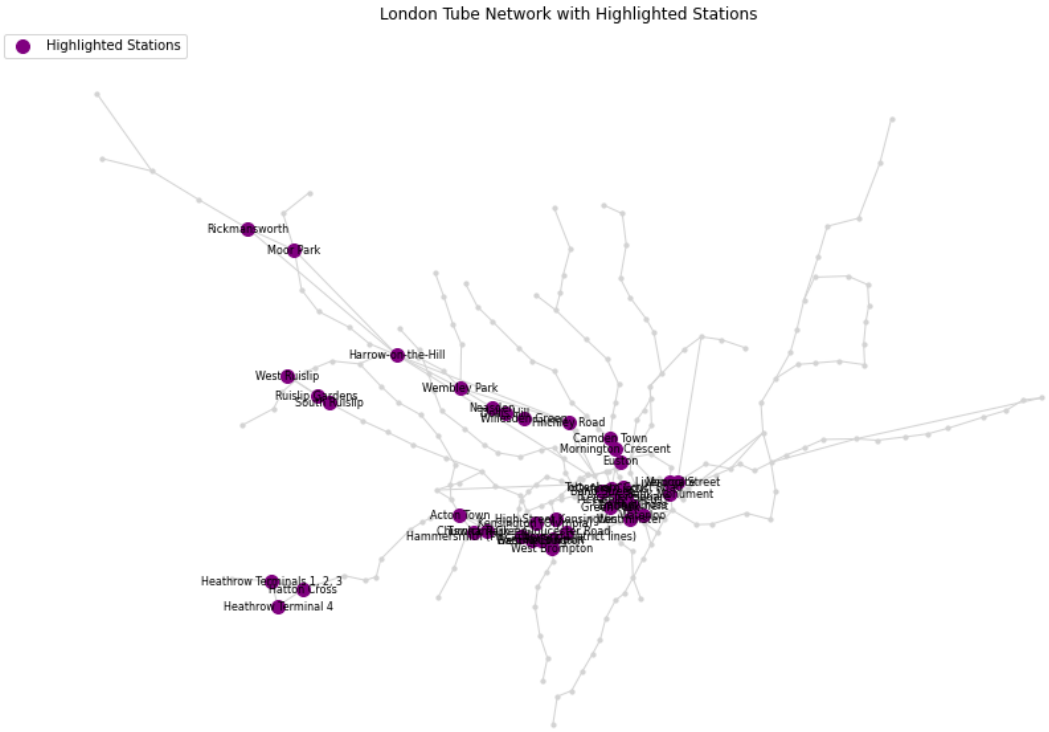
In [48]:
```
# Maximal cliques
maximal_cliques = list(nx.find_cliques(G))
clique_sizes = [len(clique) for clique in maximal_cliques]

max_clique_size = max(clique_sizes)
largest_cliques = [clique for clique in maximal_cliques if len(cliq
```

```
# Print
print(f"Number of maximal cliques: {len(largest_cliques)}")
print(f"Size of the largest clique: {max_clique_size}")
print(f"Largest clique(s): {largest_cliques}")
```

```
Number of maximal cliques: 17
Size of the largest clique: 3
Largest clique(s): [['526', '478', '463'], ['380', '333', '355'], ['
409', '282', '419'], ['502', '271', '365'], ['502', '271', '317'],
['515', '528', '344'], ['497', '440', '294'], ['423', '330', '531'],
['346', '422', '305'], ['373', '374', '372'], ['314', '406', '445'],
['445', '362', '440'], ['371', '459', '420'], ['371', '351', '518'],
['333', '520', '393'], ['333', '525', '286'], ['294', '362', '440']]
```

As previously mentioned, it is identified cliques with size 3 in the "London Underground Network" , all located in the central areas of the city and in areas of higher density like the airport London-Heathrow. These cliques reflect tightly connected subsets of stations, which are typically found in transportation hubs or central urban zones where multiple lines intersect.

In [49]:
```
nodes_in_largest_cliques = set(node for clique in largest_cliques f
plot_highlighted_stations(G, pos_geo, list(nodes_in_largest_cliques
```



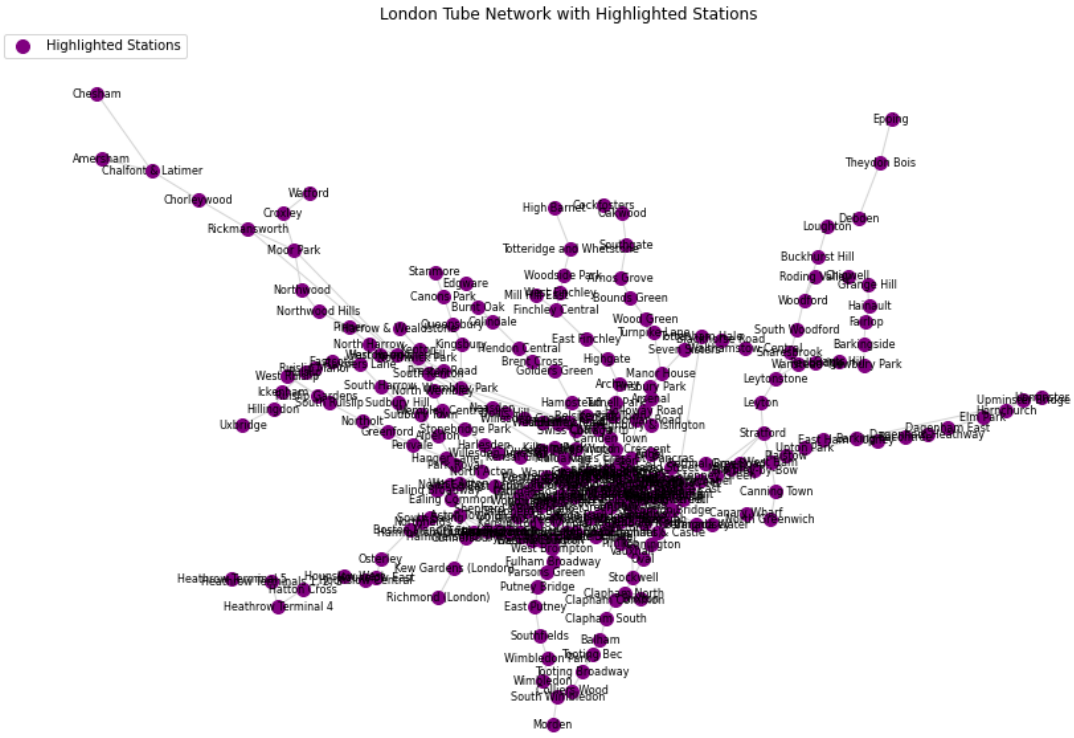
Graph Coreness (K-Cores)

When evaluating the Graph Coreness, we find as expected that only subgraphs up to k=2 can be created, in line with the tree structure of the Network. The existing stations with higher degree (hubs) are isolated as they connect to each other by lower degree stations, meaning they can not generate a subgraph. This gives a intuitive explonation of why metro systems

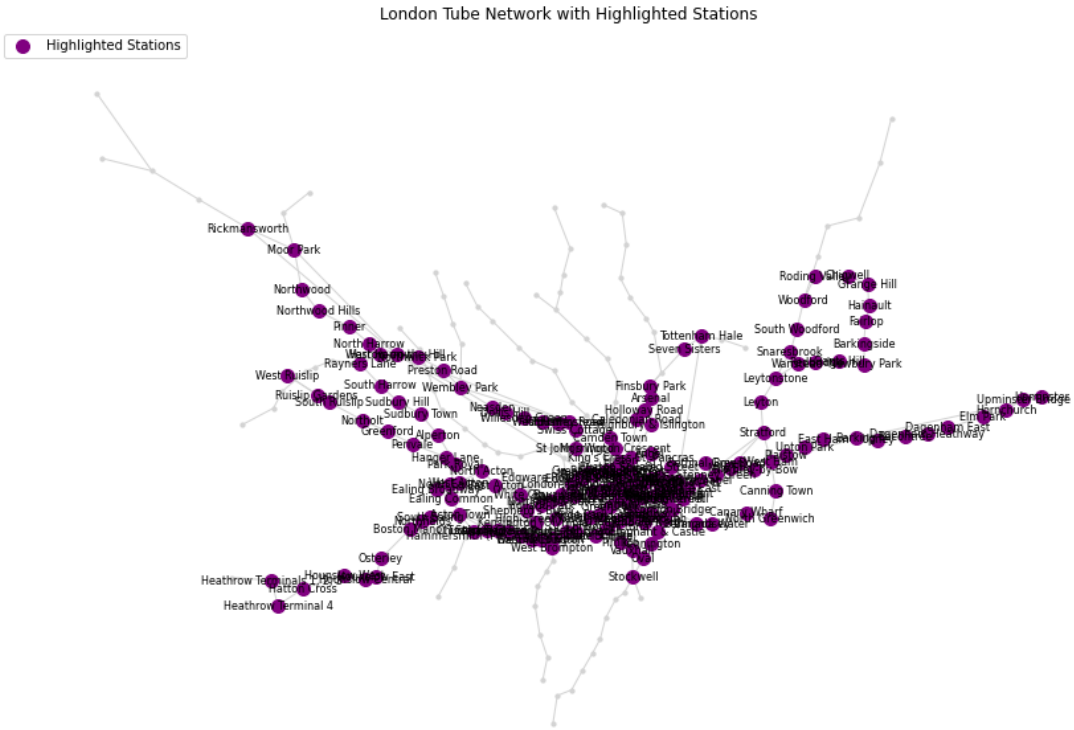
are highly sensitive to break downs, as there is a large amount of critical points that, if cut, will reduce connectivity significantly.

In [50]:
```
# Calculate the k-core for different values of k
k_core_dict = {}
for k in range(1, 4):
    k_core = nx.k_core(G, k=k)
    k_core_dict[k] = k_core

# Print the number of nodes in each k-core
print("Number of nodes in each k-core:")
for k, subgraph in k_core_dict.items():
    print(f"k = {k}: {len(subgraph.nodes())} nodes")

    # plot
    stations_kn = list(k_core_dict[k].nodes())
    plot_highlighted_stations(G, pos_geo, stations_kn, labels)
```

```
Number of nodes in each k-core:
k = 1: 268 nodes
```

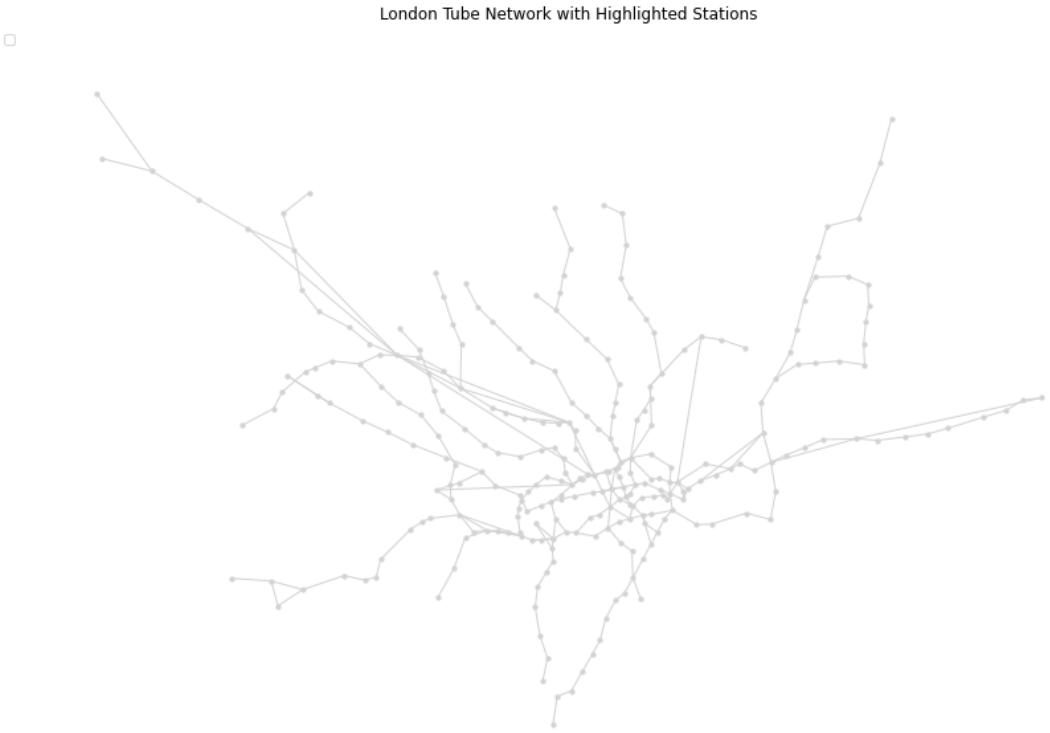



```
k = 2: 175 nodes
```




```
k = 3: 0 nodes
```

```
/var/folders/2j/t8d8dcsn1tg5sc8l4kz4_4ch0000gn/T/ipykernel_34802/1306454570.py:58: UserWarning: No artists with labels found to put in legend.  Note that artists whose label start with an underscore are ignored when legend() is called with no argument.
  plt.legend(scatterpoints=1, loc='upper left', fontsize=10)
```



Density

The the density of the "London Underground Network" is relatively small, which is in line with the tree like structure and the fact that the majority of

nodes have a degree 2, as previously discussed in relation to the adjecency Matrix. Once again, the result is directly related to the design of the transportation network where the metro lines forms the arms as linear routes, prioritizing clarity and simplicity navigation. The low density shows an small fraction of all possible connections.

In [51]:
```
# Calculate the density of the network
density = nx.density(G)
print(f"Density of the network: {density:.4f}")
```

```
Density of the network: 0.0091
```

Clustering Coefficent

Due to the network's low density, the clustering coefficient as existing triangles devided by possible triangles in the network is also very low, this is supported by the small number of observed 3-cliques (only 17 in total), indicating that fully connected triplets of stations are rare. This is in line with the low connectivity already identified.

In [52]:
```
# Calculate the global clustering coefficient (transitivity) of the
global_clustering_coefficient = nx.transitivity(G)
print(f"Global clustering coefficient of the network: {global_clust
```

```
Global clustering coefficient of the network: 0.0813
```

Small World Property

With a low clustering coefficient but a fairly long average shortest path, it can be concluded that the "London Underground Network" does not hold the small world property. This again is in line with the tree structure, meaning that many stations have to be passed to reach the outer areas of the metro network, with no direct connections available. This might however change if additional transportation networks like regional trains or express busses would be considered, which may provide cross-connections that reduce travel distances and increase local clustering, bringing the network closer to small-world behavior.

In [53]:
```
actual_avg_shortest_path_length = nx.average_shortest_path_length(G
print(f"Average Shortest Path Length: {actual_avg_shortest_path_len
```

```
Average Shortest Path Length: 12.4380
```

Community Detection

The objective of the following section is to define the optimal graph partition. This will be measured using modularity, which quantifies the quality of the partition. A higher modularity value indicates a better partition, with values approaching 1 representing strong community structure, while values closer to

-1 suggest poor or weak partitions.

Optimal Modularity: *Alternatively* **Greddy Modularity Algorithm**

As previously mentioned, the goal is to identify the partition that maximizes modularity. The following code implements the optimal algorithm. However, due to the size of the "London Underground Network", running this algorithm has an exponential time complexity, making it computationally expensive.

In [54]:
```
# #If G is a networkx graph
# G_nx = G

# #Convert to igraph
# G_ig = ig.Graph.TupleList(G_nx.edges(), directed=False)

# #Run community detection
# clusters = G_ig.community_optimal_modularity()

# #rint community membership
# print(clusters.membership)
```

As previously mentioned, the goal is to identify the partition that maximizes modularity. The following code implements the optimal algorithm. However, due to the size of the "London Underground Network", running this algorithm has an exponential time complexity, making it computationally expensive.

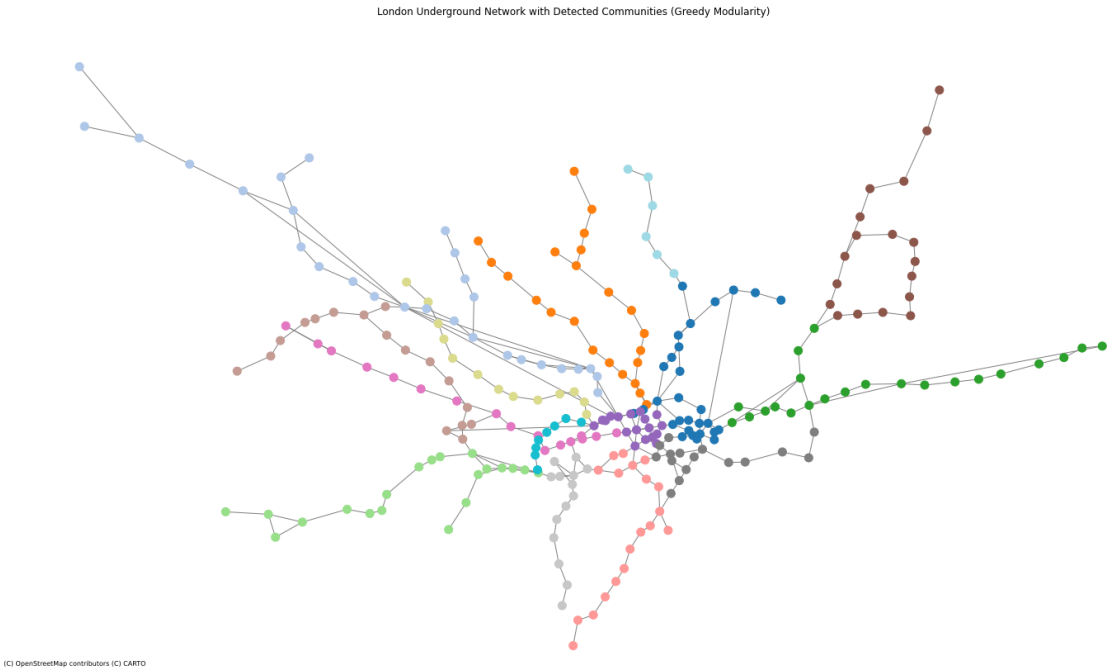
Therefore, the next algorithm to be used is Greedy Modularity, which aims to efficiently approximate the division of the network into communities by maximizing modularity in a computationally feasible manner. Although it does not guarantee the same result as the optimal modularity algorithm, it is suitable for large networks like ours and provides an acceptable level of accuracy.

The algorithm detected 15 communities, a reasonable number considering that the "London Underground Network" is based on metro lines. As mentioned in the introduction, the network consists of 11 metro lines, as well as other lines (such as Light Rail and Regional Lines). The modularity value of 0.8192 further suggests a good construction of communities within the network.

Interestingly, when analyzing the graph, we observe that the main difference between the communities and the real metro lines is that, once reaching the center of the map, the communities diverge into different groups. This contrasts with the real metro lines, which extend from one extreme of the city to the other. This is an intriguing feature of the algorithm, as it identifies communities that are not necessarily aligned with the physical metro line structure.

In [55]:
```
# Compute communities using greedy modularity
communities = list(greedy_modularity_communities(G))

community_mapping = {node: idx for idx, community in enumerate(comm
nx.set_node_attributes(G, community_mapping, 'community')

# Print
print(f"Number of communities detected: {len(communities)}")
print(f"Modularity: {modularity(G, communities):.4f}")


# Plot
colors = [community_mapping[node] for node in G.nodes()]
fig, ax = plt.subplots(figsize=(25, 25))
ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

nx.draw(G, pos=pos_geo, with_labels=False, node_size=100,
        node_color=colors, cmap=plt.cm.tab20, edge_color='gray', ax

ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network with Detected Communities (Gr
plt.show()
```

```
Number of communities detected: 15
Modularity: 0.8192
```



Fast Greedy Algorithm

To simplify the integer programming proposed by the optimal modularity algorithm, which cannot be run on our network due to its computational complexity, the Fast Greedy algorithm will be applied to the "London Underground Network".

In this case, the algorithm yields a slightly better modularity value than the

previous Greedy Modularity algorithm. However, both algorithms converge to detect 15 communities, with a similar pattern in the layout map. The communities are primarily located in the parts of the metro lines outside the city center, while the stations in the city center form distinct communities, possibly based on their connectivity to other stations.

In [56]:
```
# Convert to iGraph
edges = list(G.edges())
ig_graph = ig.Graph.TupleList(edges, directed=False)

# Run Fast Greedy community detection
communities = ig_graph.community_fastgreedy()
clusters = communities.as_clustering()

ig_to_nx = {i: v["name"] for i, v in enumerate(ig_graph.vs)}
nx_community_mapping = {ig_to_nx[i]: clusters.membership[i] for i i
nx.set_node_attributes(G, nx_community_mapping, 'community')

# Print
print(f"Number of communities detected: {len(clusters)}")
print(f"Modularity: {clusters.modularity:.4f}")

# Plot over map using pos_geo and community colors
colors = [nx_community_mapping[node] for node in G.nodes()]

fig, ax = plt.subplots(figsize=(25, 25))
ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

nx.draw(
    G, pos=pos_geo, with_labels=False, node_size=100,
    node_color=colors, cmap=plt.cm.tab20, edge_color='gray', ax=ax
)

ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network with Fast Greedy Communities
plt.show()
```

```
Number of communities detected: 15
Modularity: 0.8196
```

Louvain Algorithm

As an alternative, the Louvain Algorithm can also be used to find the partition that maximizes modularity without the need to run an optimization algorithm.

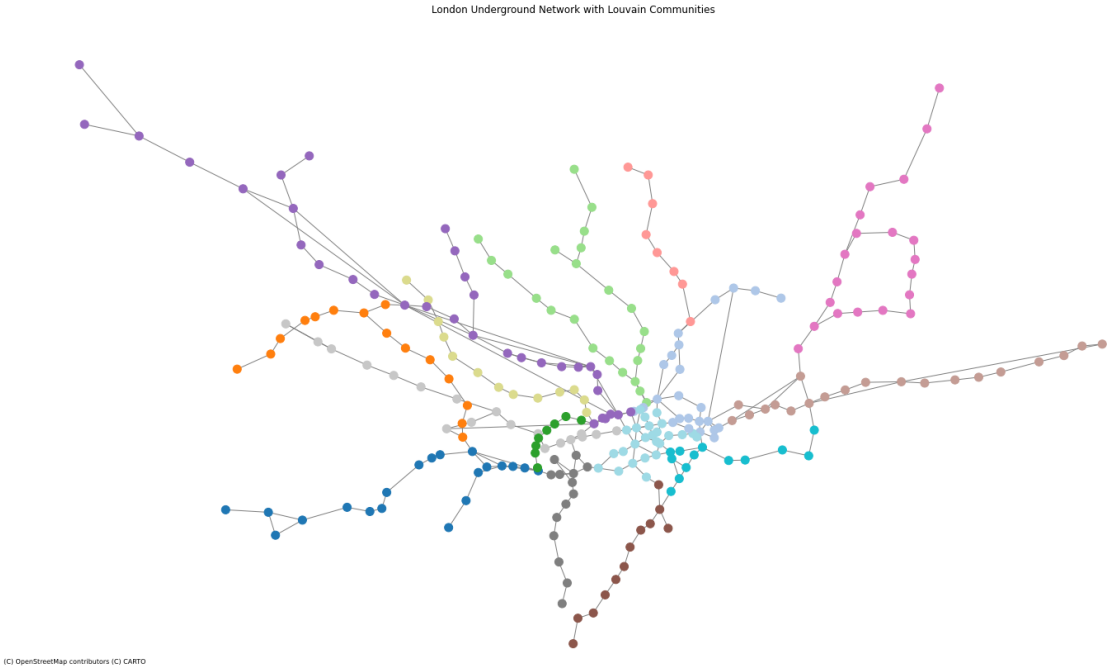
Once again, the Louvain algorithm detects 15 communities, consistent with the results from the previous two algorithms. However, the modularity increases to 0.823, which is a positive indicator of the algorithm's ability to accurately detect communities. This value surpasses the modularity obtained by the Greedy Modularity and Fast Greedy algorithms.

Similarly, the communities identified by the Louvain algorithm appear to represent the "arms" of the network. These can be interpreted as the metro lines or stations located in the outer areas of the city center. Within the city center, one to three communities are identified, possibly based on their connectivity to key junction stations that serve as transfer points for passengers.

In [57]:
```
# Perform Louvain community detection
partition = community_louvain.best_partition(G, weight='weight')  #
nx.set_node_attributes(G, partition, 'community')

# Print
num_communities = len(set(partition.values()))
print(f"Number of communities detected: {num_communities}")

# Compute modularity
modularity_value = community_louvain.modularity(partition, G, weigh
print(f"Modularity of the detected community structure: {modularity_

# Plot on London map
community_colors = [partition[node] for node in G.nodes()]
fig, ax = plt.subplots(figsize=(25, 25))
```

```
ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

nx.draw(
    G,
    pos=pos_geo,
    node_color=community_colors,
    with_labels=False,
    node_size=100,
    cmap=plt.cm.tab20,
    edge_color='gray',
    ax=ax
)

ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network with Louvain Communities")
plt.show()
```

```
Number of communities detected: 15
Modularity of the detected community structure: 0.8268
```



Label Propagation Algorithm

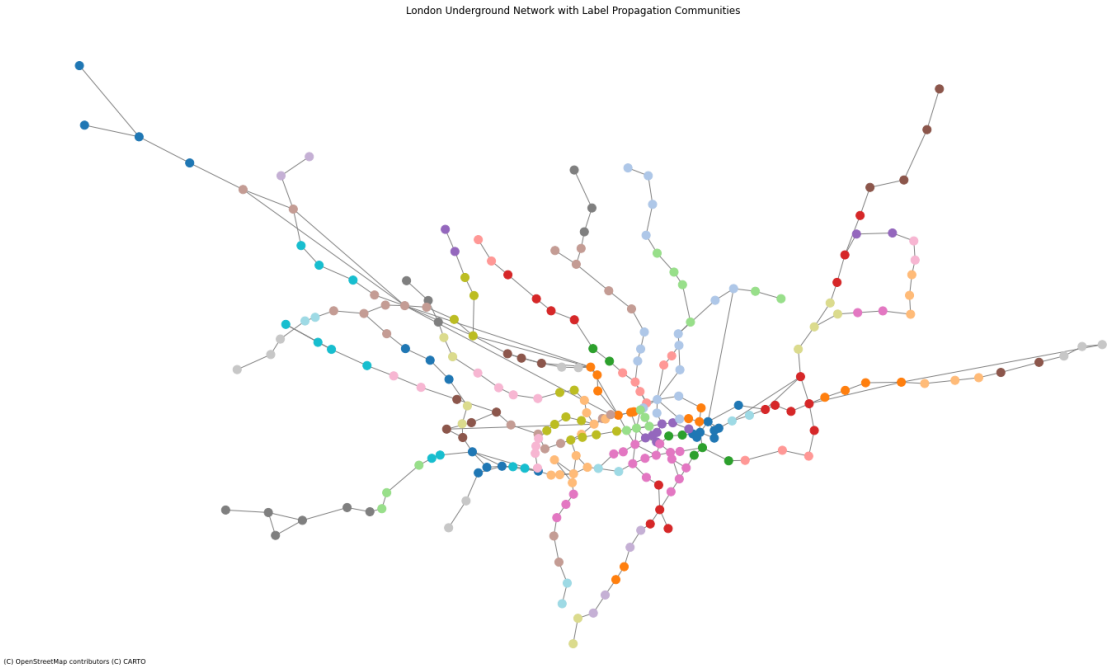
The Label Propagation Algorithm takes into account the neighbors of each vertex for community detection, making it a useful option for large networks, such as the one in this analysis of the "London Underground Network".

The result of applying this algorithm yields 77 communities with a significantly reduced modularity of 0.62, which indicates worse community detection compared to the previous algorithms. Additionally, the large number of communities, 77, is not well-defined, as there are no clear breaks in the network that would correspond to such a fragmented division of metro lines. This suggests that the algorithm is identifying communities based on smaller

groups of stations, with each community containing roughly 4 stations. This is likely due to the algorithm's tendency to over-segment the network, possibly focusing on local neighborhoods in the city rather than larger areas of London.

In [58]:
```
# Detect communities using Label Propagation
communities = list(label_propagation_communities(G))

community_mapping = {node: idx for idx, community in enumerate(comm
nx.set_node_attributes(G, community_mapping, 'community')

# Print results
print(f"Number of communities detected: {len(communities)}")

# Compute modularity
modularity_value = modularity(G, communities)
print(f"Modularity of the detected community structure: {modularity_

# Plot
fig, ax = plt.subplots(figsize=(25, 25))
ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

community_colors = [community_mapping[node] for node in G.nodes()]


nx.draw(
    G,
    pos=pos_geo,
    node_color=community_colors,
    with_labels=False,
    node_size=100,
    cmap=plt.cm.tab20,
    edge_color='gray',
    ax=ax
)

ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network with Label Propagation Commun
plt.show()
```

```
Number of communities detected: 77
Modularity of the detected community structure: 0.6271
```



Edge Betweenness Algorithm

The Edge Betweenness Algorithm results in 2 communities in the network, with a much lower modularity of 0.4194. This is not optimal when compared to the results of the other algorithms. Additionally, the communities identified are quite large and seem to correspond to a division between the South West and North East parts of the network, according to the map layout.

This result stems from the algorithm's method of creating communities by iteratively removing central edges. As determined by the earlier centrality measures, these central edges are predominantly located in downtown London. These edges have the highest edge betweenness centrality. In contrast to other algorithms, which identified this central area as being split into one to three communities, the Edge Betweenness Algorithm focuses on this central region as a point of separation, resulting in the identified two large communities.

In [59]:
```
# Run Girvan-Newman and get first level of split
communities_generator = girvan_newman(G)
first_level_communities = next(communities_generator)
communities = [set(community) for community in first_level_communit

community_mapping = {node: idx for idx, community in enumerate(comm
nx.set_node_attributes(G, community_mapping, 'community')

# Print number of communities and modularity
print(f"Number of communities detected: {len(communities)}")

modularity_value = modularity(G, communities)
print(f"Modularity of the detected community structure: {modularity_

# Plot over London map
fig, ax = plt.subplots(figsize=(25, 25))
```

```
ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

community_colors = [community_mapping[node] for node in G.nodes()]


nx.draw(
    G,
    pos=pos_geo,
    node_color=community_colors,
    with_labels=False,
    node_size=100,
    cmap=plt.cm.tab20,
    edge_color='gray',
    ax=ax
)

ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network with Girvan–Newman Communitie
plt.show()
```

```
Number of communities detected: 2
Modularity of the detected community structure: 0.4194
```

Walktrap Algorithm

The Walktrap Algorithm detects communities by simulating a large number of random walks between vertices. In this case, the algorithm identifies 33 communities with a modularity of 0.77, which is an acceptable value but still lower than the modularity found in previous algorithms.

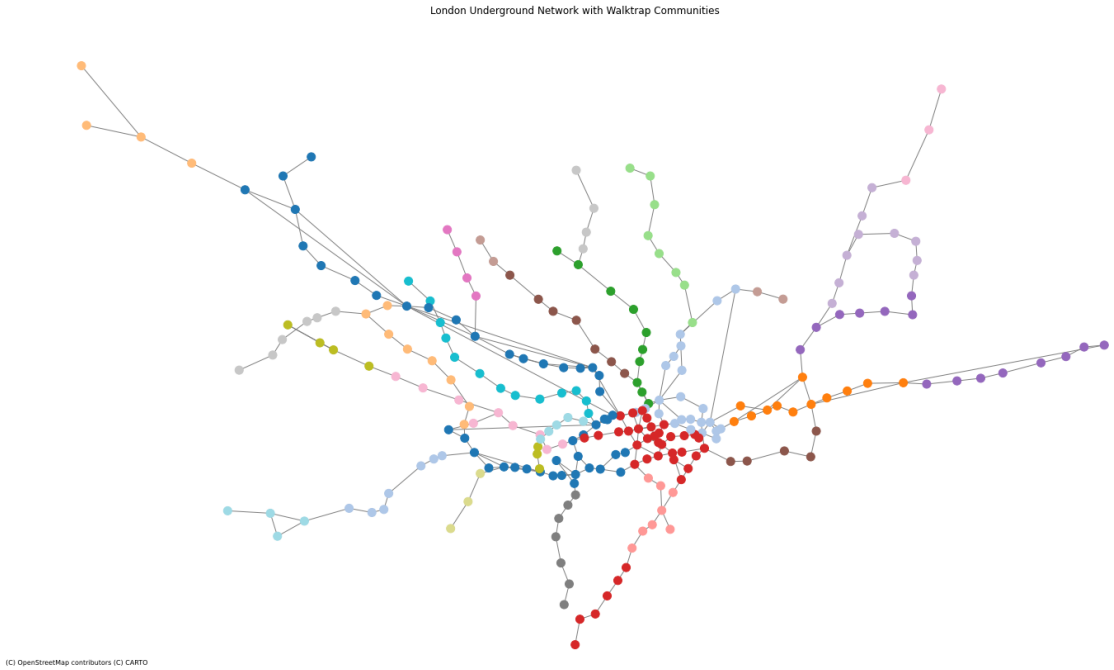
The relatively high number of communities can be attributed to the algorithm's tendency to create clusters with small sizes. This is a characteristic feature of the Walktrap algorithm, which tends to identify finer-grained communities

rather than larger, more broadly defined ones, unlike the Label Propagation Algorithm which resulted in a much higher number of communities. While the communities detected here are smaller and more numerous, they are not as detailed as those produced by the Label Propagation Algorithm.

One interesting aspect of the Walktrap Algorithm is its reliance on random walks, which simulate the likelihood of vertices being part of the same community based on their proximity to each other in the graph, also it allows the algorithm to capture finer distinctions between groups.

In [60]:
```
# Run Walktrap community detection
walktrap_communities = ig_graph.community_walktrap(weights=None)
clusters = walktrap_communities.as_clustering()

ig_to_nx = {i: v["name"] for i, v in enumerate(ig_graph.vs)}
nx_community_mapping = {ig_to_nx[i]: clusters.membership[i] for i i
nx.set_node_attributes(G, nx_community_mapping, 'community')

# Format communities as list of sets for modularity calculation
num_communities = len(clusters)
print(f"Number of communities detected: {num_communities}")
communities = [[] for _ in range(num_communities)]
for node, com_id in nx_community_mapping.items():
    communities[com_id].append(node)
communities = [set(c) for c in communities]

# Compute modularity
modularity_value = modularity(G, communities)
print(f"Modularity of the detected community structure: {modularity_

# Plot on London map
fig, ax = plt.subplots(figsize=(25, 25))
ctx.add_basemap(ax, crs="EPSG:4326", source=ctx.providers.CartoDB.P

community_colors = [nx_community_mapping[node] for node in G.nodes(

nx.draw(
    G,
    pos=pos_geo,
    node_color=community_colors,
    with_labels=False,
    node_size=100,
    cmap=plt.cm.tab20,
    edge_color='gray',
    ax=ax
)

ax.set_xlim(-0.68, 0.29)
ax.set_ylim(51.39, 51.73)
ax.set_aspect(1 / np.cos(np.radians((51.4 + 51.73) / 2)))

plt.title("London Underground Network with Walktrap Communities")
plt.show()
```

```
Number of communities detected: 33
Modularity of the detected community structure: 0.7719
```



Summary Community Detection Methods

After running several community detection algorithms, the resulting communities were visually reviewed to assess their characteristics. The first three algorithms listed in the table below all identified the same number of communities, 15, which is a reasonable figure given the number of metro lines in the "London Underground Network" and its overall structure.

Among these, the Louvain algorithm produced the highest modularity value (0.8235),by a small margin, indicating the most optimal partitioning of the network. This high modularity suggests a strong alignment between the detected communities and the underlying structure of the network.

The layout maps produced by these three algorithms reveal that the detected communities largely correspond to the "arms" of the network metro lines extending outward from the city center. These algorithms effectively capture the tree-like structure of the network, identifying each arm as a distinct community. However, they fall short in representing the full extent of individual metro lines; instead of grouping opposite arms of the same line into a single community, they treat each arm as a separate entity. Despite this limitation, the algorithms demonstrate a good ability to capture the core structure of the network.

Additionally, communities in the city center represent major interchanging metro stations, which act as key transfer points for passengers. This partitioning reflects a more natural and practical division of the network, aligning with real-world patterns of connectivity and transit usage.

In contrast, the remaining three algorithms struggled to effectively model the network's tree-like structure. They produced inconsistent community counts, ranging from as few as 2 to as many as 77, indicating difficulty in identifying a coherent partitioning according to the metro lines in the transportation network.

Algorithm	Communities	Modularity
Greedy Modularity Algorithm	15	0.8192
Fast Greedy Algorithm	15	0.8196
Louvain Algorithm	**15**	**0.8235**
Label Propagation Algorithm	77	0.6271
Edge Betweenness Algorithm	2	0.4194
Walktrap Algorithm	33	0.7719

Assortativity

The degree assortativity in the "London Underground Network" is low, meaning that high-degree nodes (such as metros stations in the city center) tend to connect more frequently to low-degree nodes rather than other high-degree nodes. As mentioned repetitively during the analysis, most of the results coe from the tree like structure of the transportation network, where is focused on central stations a a point to distribute the passenger traffic to other branches or metro lines, that in the network will be the metro lines. Rather than form an dense network as a dense web, the arms where represents the peripheral metro lines aims for the efficiency and avoid the redundancy in the reaching points of the city throught them.

In [61]:
```
# Calculate the assortativity coefficient of the network
assortativity_coefficient = nx.degree_assortativity_coefficient(G)
print(f"Assortativity Coefficient of the network: {assortativity_co
```

```
Assortativity Coefficient of the network: 0.1679
```

Models and Inference

The following section contains network models like random graphs in an attempt to extract characteristics and do inference for the network at hand

Random Graphs

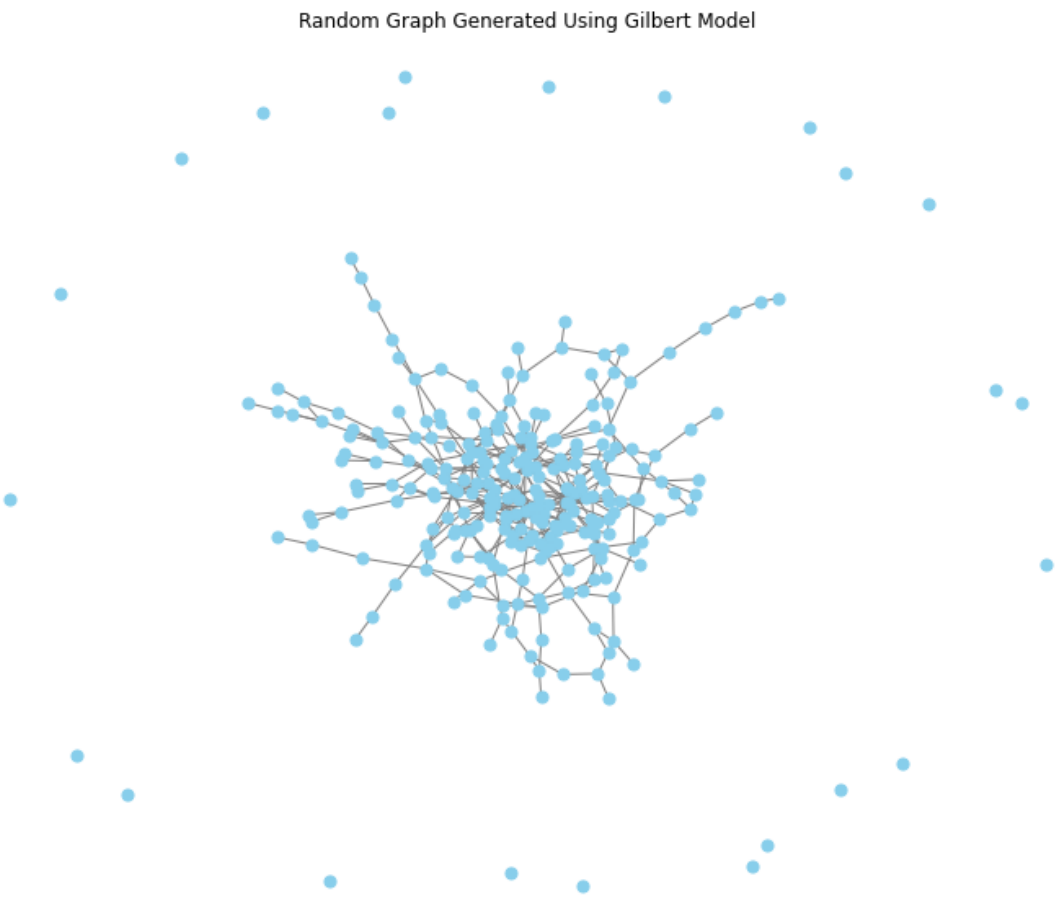
We start by generating three random graphs, with two from the set of generalized random graphs:

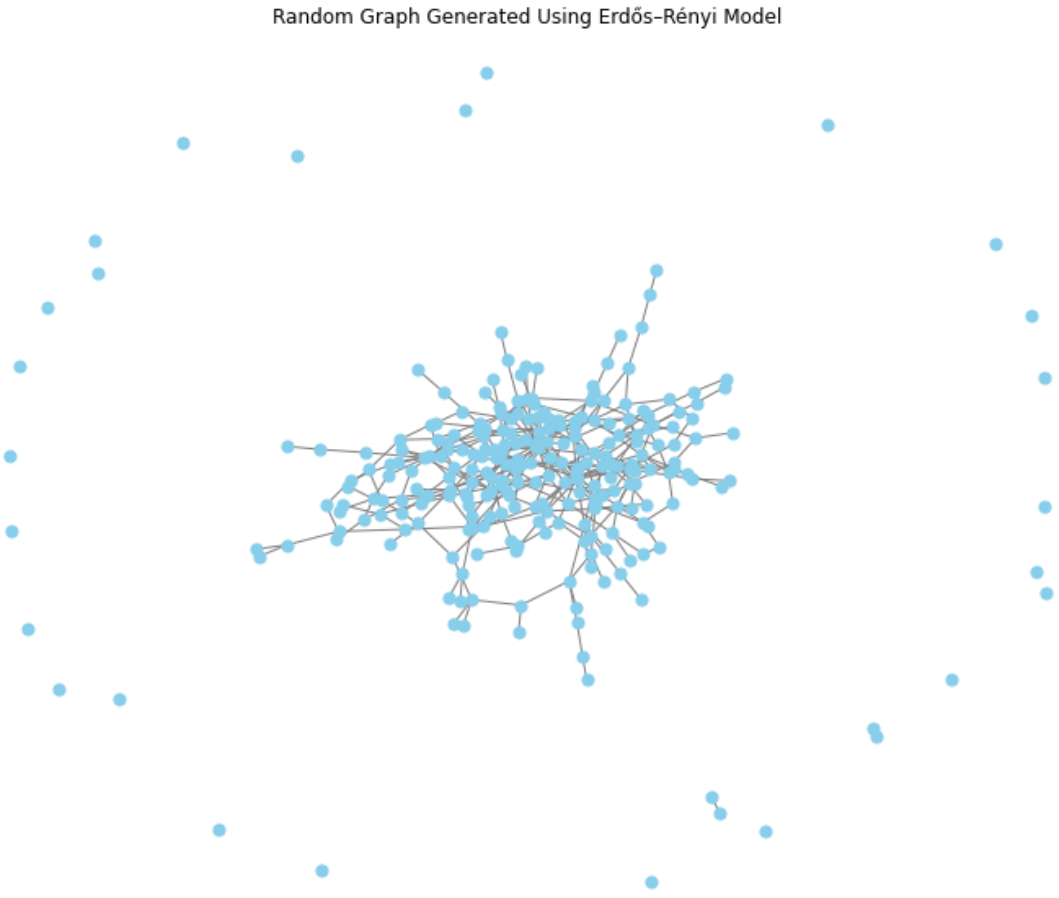
- Gilbert Model
- Erdős–Rényi Random Graph
- Configuration Random Graph

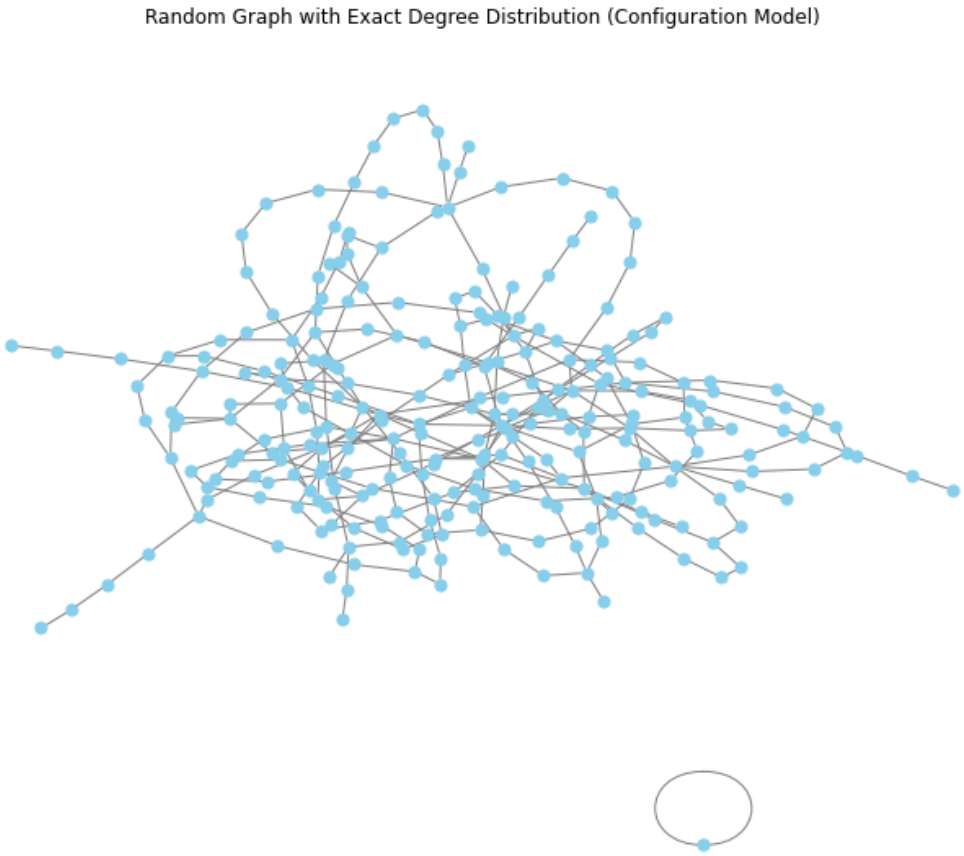
Plotting both graphs via spring layout, we see that the first two show a large number of isolated nodes. Beyond that, only he Configuration Graph shows a structure remotely similar to the original network. This is not unexpected, as the first two give equal probabilities to the connections between any two nodes, while the configuration graph has a fixed degree distribution and thus is forced into developing somewhat of a tree structure. It shows however how inprobable it is to generate the kind of metro network we have randomly.

In [69]:
```
# Generate random Graphs
random.seed(42)

num_nodes = G.number_of_nodes()
num_edges = G.number_of_edges()
p = num_edges / (num_nodes * (num_nodes - 1) / 2)  # Probability of

# Gilbert
gilbert_random_graph = nx.erdos_renyi_graph(n=num_nodes, p=p, seed=

# Erdős–Rényi
erdos_renyi_random_graph = nx.gnm_random_graph(n=num_nodes, m=num_e

# Configuration model
degree_sequence = [deg for _, deg in G.degree()]
configuration_model_graph = nx.configuration_model(degree_sequence)

# Plot
fig, axes = plt.subplots(3, 1, figsize=(10, 24))  # Adjust layout t

# Gilbert
pos_gilbert = nx.spring_layout(gilbert_random_graph, seed=42)  # Us
nx.draw(
    gilbert_random_graph, pos_gilbert, with_labels=False, node_size
    node_color='skyblue', edge_color='gray', ax=axes[0]
)
axes[0].set_title("Random Graph Generated Using Gilbert Model")

# Erdős–Rényi
pos_erdos = nx.spring_layout(erdos_renyi_random_graph, seed=42)  # (
nx.draw(
    erdos_renyi_random_graph, pos_erdos, with_labels=False, node_si
    node_color='skyblue', edge_color='gray', ax=axes[1]
)
axes[1].set_title("Random Graph Generated Using Erdős–Rényi Model")

# Configuration Model
pos_config = nx.spring_layout(configuration_model_graph, seed=42)
nx.draw(
    configuration_model_graph, pos_config, with_labels=False, node_
    node_color='skyblue', edge_color='gray', ax=axes[2]
```

```
)
axes[2].set_title("Random Graph with Exact Degree Distribution (Con

plt.tight_layout()
plt.show()
```





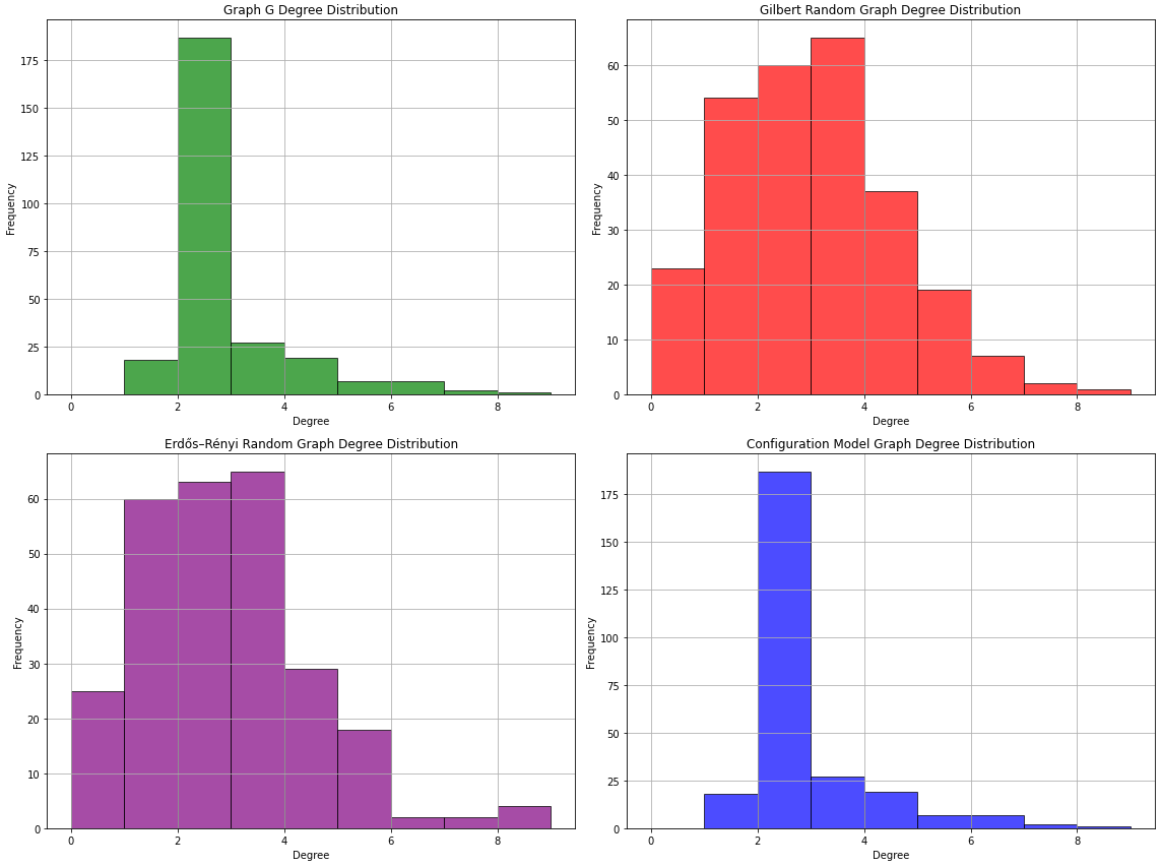


The observations from above are validated via the degree distributions, as both the Gilbert Model and the Erdös-Renyi show significantly different degree distributions due to the independence of generated connections. Showing the sum of all degrees, we find all models in similar ranges again, thus due to the set up of the graph generation the distribution shifts.

In [70]:
```
### Degree Distribution

# Plot
plt.figure(figsize=(16, 12))

#original graph
plt.subplot(2, 2, 1)
plt.hist(degree_sequence, bins=range(0, 8 + 2),
         edgecolor='black', alpha=0.7, color='green')
plt.title('Graph G Degree Distribution')
plt.xlabel('Degree')
plt.ylabel('Frequency')
plt.grid(True)

#Gilbert random graph
plt.subplot(2, 2, 2)
plt.hist([deg for _, deg in gilbert_random_graph.degree()],
         bins=range(0, 8 + 2),
         edgecolor='black', alpha=0.7, color='red')
plt.title('Gilbert Random Graph Degree Distribution')
plt.xlabel('Degree')
```

```
plt.ylabel('Frequency')
plt.grid(True)

#Erdős–Rényi random graph
plt.subplot(2, 2, 3)
plt.hist([deg for _, deg in erdos_renyi_random_graph.degree()],
         bins=range(0, 8 + 2),
         edgecolor='black', alpha=0.7, color='purple')
plt.title('Erdős–Rényi Random Graph Degree Distribution')
plt.xlabel('Degree')
plt.ylabel('Frequency')
plt.grid(True)

#Configuration Model random graph
plt.subplot(2, 2, 4)
plt.hist([deg for _, deg in configuration_model_graph.degree()], bi
         edgecolor='black', alpha=0.7, color='blue')
plt.title('Configuration Model Graph Degree Distribution')
plt.xlabel('Degree')
plt.ylabel('Frequency')
plt.grid(True)


# Calculate the sum of degrees for all networks
original_graph_degree_sum = sum(degree_sequence)
gilbert_graph_degree_sum = sum(deg for _, deg in gilbert_random_gra
erdos_renyi_graph_degree_sum = sum(deg for _, deg in erdos_renyi_ra
configuration_model_graph_degree_sum = sum(deg for _, deg in config

# Print the results
print(f"Sum of Degrees - Original Graph: {original_graph_degree_sum
print(f"Sum of Degrees - Gilbert Random Graph: {gilbert_graph_degre
print(f"Sum of Degrees - Erdős–Rényi Random Graph: {erdos_renyi_gra
print(f"Sum of Degrees - Configuration Model Graph: {configuration_

plt.tight_layout()
plt.show()
```

```
Sum of Degrees - Original Graph: 648
Sum of Degrees - Gilbert Random Graph: 676
Sum of Degrees - Erdős–Rényi Random Graph: 648
Sum of Degrees - Configuration Model Graph: 648
```



We then run louvain community detection (as the best results yielding algortihm for out network) on the random graphs and unsurprisingly find a large number of communties for the first two, in part due to the number os isolated nodes. For the configuration model, the same ammount of communties as for the original network is found and the tree arms are somewhat well recognized, even though overall modularity is lower. This is expected as the random graph is not fully able to represent the tree like structure which led to the high modularity previously.

In [71]:
```
def apply_louvain_and_plot(random_graph_1, random_graph_2, random_g

    # Louvain community detection
    partition_1 = community_louvain.best_partition(random_graph_1)
    modularity_1 = community_louvain.modularity(partition_1, random
    colors_1 = [partition_1[node] for node in random_graph_1.nodes(

    partition_2 = community_louvain.best_partition(random_graph_2)
    modularity_2 = community_louvain.modularity(partition_2, random
    colors_2 = [partition_2[node] for node in random_graph_2.nodes(

    partition_3 = community_louvain.best_partition(random_graph_3)
    modularity_3 = community_louvain.modularity(partition_3, random
    colors_3 = [partition_3[node] for node in random_graph_3.nodes(

    # Plot
    fig, axes = plt.subplots(nrows=3, ncols=1, figsize=(8, 24))  # /

    pos_1 = nx.spring_layout(random_graph_1, seed=42)
    nx.draw(
```

```
        random_graph_1, pos_1, node_color=colors_1, with_labels=Fal
        node_size=50, cmap=plt.cm.tab20, edge_color='gray', ax=axes
    )
    axes[0].set_title(f"Gilbert Model (Modularity: {modularity_1:.4

    pos_2 = nx.spring_layout(random_graph_2, seed=42)
    nx.draw(
        random_graph_2, pos_2, node_color=colors_2, with_labels=Fal
        node_size=50, cmap=plt.cm.tab20, edge_color='gray', ax=axes
    )
    axes[1].set_title(f"Erdös Renyi (Modularity: {modularity_2:.4f}

    pos_3 = nx.spring_layout(random_graph_3, seed=42)
    nx.draw(
        random_graph_3, pos_3, node_color=colors_3, with_labels=Fal
        node_size=50, cmap=plt.cm.tab20, edge_color='gray', ax=axes
    )
    axes[2].set_title(f"Configuration Model (Modularity: {modularit

    plt.tight_layout()
    plt.show()

# Apply to random graphs
apply_louvain_and_plot(gilbert_random_graph, erdos_renyi_random_gra
```

Erdös Renyi (Modularity: 0.6699, Communities: 42)

Configuration Model (Modularity: 0.7454, Communities: 16)

In conclusion, it is very apparent that random graphs struggle with the kind of extreme tree like structure the metro network exhibits. Only by fixing explicitly the degree distribution we yield a graph somewhat similar, but even then the extent of the roots of the tree is not modeled well, due to a very low probability of generating arms with the length found in the original network.

Small World Property

Unsurprisingly, the generated configuration model Random Graph does not fulfill the the small-world property, as the orginial graph does not fulfill it. We however only compare the configuration graph, as the Gilbert model Graph has so many isolated nodes that the average shortest path becomes unrepresentative. For the configuration model, we find that the resulting average shortest path is shorter than that of the original graph, in line with the observed problems with its ability to construct the tree like structure found in the original graph. Meanwhile the clustering decreased, indicating a lower average density. While antiintuitive at first, this is in line with an equal lack in density at the Network center, where the original graph showed a very high density. The attributes of the original graph thus seem to have been smoothed out in a way.

In [72]:
```
# Original Graph
G.remove_nodes_from(list(nx.isolates(G)))
original_clustering = nx.average_clustering(G)
original_avg_shortest_path = nx.average_shortest_path_length(G)

print("Original Graph:")
print(f"  Clustering Coefficient: {original_clustering:.4f}")
print(f"  Average Shortest Path Length: {original_avg_shortest_path

# configuration model Graph
configuration_model_graph = nx.Graph(configuration_model_graph)  # (
configuration_model_graph.remove_edges_from(nx.selfloop_edges(confi(
configuration_model_graph.remove_nodes_from(list(nx.isolates(config(
config_clustering = nx.average_clustering(configuration_model_graph
config_avg_shortest_path = nx.average_shortest_path_length(configur(

print("\nConfiguration Model Graph:")
print(f"  Clustering Coefficient: {config_clustering:.4f}")
print(f"  Average Shortest Path Length: {config_avg_shortest_path:.(
```

```
Original Graph:
  Clustering Coefficient: 0.0555
  Average Shortest Path Length: 12.4380

Configuration Model Graph:
  Clustering Coefficient: 0.0017
  Average Shortest Path Length: 7.5982
```

Community detection

The Louvain algorithm detected 35 communities in the Gilbert (Erdős–Rényi) random graph and 16 communities in the configuration model graph. This suggests that while the Gilbert graph, led to a large number of small and more detailed communities, as seem before in previous algorithm can lead to geographical neighbours in the city threated as an indivual communities, in the other hand the configuration model's preserved degree distribution resulting in fewer and more meaningful groupings, but what is more important close to the specified communities corresponding to the metro lines which are 11 idientified.

In [73]:
```
import community.community_louvain as community_louvain

partition_1 = community_louvain.best_partition(gilbert_random_graph
partition_2 = community_louvain.best_partition(configuration_model_(

# Count number of communities
num_communities_1 = len(set(partition_1.values()))
num_communities_2 = len(set(partition_2.values()))

print(f"Gilbert random graph - Communities detected: {num_communiti(
print(f"Configuration model graph - Communities detected: {num_commu
```

```
Gilbert random graph - Communities detected: 37
Configuration model graph - Communities detected: 15
```

In the **Fast Greedy** analysis of community structure revealed some interesting insights into the network's modularity. The Gilbert random graph, with its probability based edge formation, yielded 9 detected communities, while the configuration model, (with each node having a degree of 2, that was found earlier as the predominant value in the nodes), resulted in 10 detected communities, pretty close to the 11 metro lines. The small difference in the number of communities suggests that, while both models share similar overall structure, the degree sequence constraint in the Configuration Model might lead to a slightly more real metro lines communities rather thant the randomly generated.

In [74]:
```
import igraph as ig

random.seed(42)

# Create a Gilbert (Erdős–Rényi) random graph
```

```
g_er = ig.Graph.Erdos_Renyi(n=100, p=0.05)

# Create a Configuration Model random graph
degree_sequence = [2] * 100  #
g_config = ig.Graph.Degree_Sequence(degree_sequence, method="vl")


# Fast Greedy on ER graph
er_dendrogram = g_er.community_fastgreedy()
er_clusters = er_dendrogram.as_clustering()
num_communities_er = len(er_clusters)

# Fast Greedy on Configuration Model
config_dendrogram = g_config.community_fastgreedy()
config_clusters = config_dendrogram.as_clustering()
num_communities_config = len(config_clusters)

print(f"Gilbert random graph - Communities detected: {num_communiti
print(f"Configuration model graph - Communities detected: {num_comm
```

```
Gilbert random graph - Communities detected: 8
Configuration model graph - Communities detected: 10
```

In [75]:
```
# Create a Gilbert (Erdős–Rényi) random graph
g_er = ig.Graph.Erdos_Renyi(n=100, p=0.05)

# Create a Configuration Model random graph
degree_sequence = [3] * 100  # Regular degree sequence
g_config = ig.Graph.Degree_Sequence(degree_sequence, method="vl")

# Label Propagation on ER graph
er_communities = g_er.community_label_propagation()
num_communities_er = len(er_communities)

# Label Propagation on Configuration Model
config_communities = g_config.community_label_propagation()
num_communities_config = len(config_communities)

print(f"Gilbert random graph - Communities detected: {num_communiti
print(f"Configuration model graph - Communities detected: {num_comm
```

```
Gilbert random graph - Communities detected: 1
Configuration model graph - Communities detected: 9
```

In [76]:
```
# import igraph as ig
# import networkx as nx
# import numpy as np
# import random
# from tqdm import tqdm  # Optional: for progress bar

# # Set seeds for reproducibility
# random.seed(10)
# np.random.seed(10)

# # Use the igraph graph (ig_graph) you've already defined
# N = ig_graph.vcount()
# L = ig_graph.ecount()
```

```
# n_trials = 10000

# # Preallocate arrays
# num_comm_ER_fg = np.zeros(n_trials)
# num_comm_ER_lo = np.zeros(n_trials)
# num_comm_ER_lp = np.zeros(n_trials)
# num_comm_ER_eb = np.zeros(n_trials)
# num_comm_ER_wa = np.zeros(n_trials)

# for i in tqdm(range(n_trials), desc="Running community detection (
#     # Generate an Erdős–Rényi graph with same N and L
#     ER_graph = ig.Graph.Erdos_Renyi(n=N, m=L)

#     # Fast Greedy
#     cl_fg = ER_graph.community_fastgreedy().as_clustering()
#     num_comm_ER_fg[i] = len(cl_fg)

#     # Louvain
#     cl_lo = ER_graph.community_multilevel()
#     num_comm_ER_lo[i] = len(cl_lo)

#     # Label Propagation (optional: use Louvain membership as init
#     cl_lp = ER_graph.community_label_propagation(initial=cl_lo.me
#     num_comm_ER_lp[i] = len(cl_lp)

#     # Edge Betweenness
#     cl_eb = ER_graph.community_edge_betweenness(directed=False).a
#     num_comm_ER_eb[i] = len(cl_eb)

#     # Walktrap
#     cl_wa = ER_graph.community_walktrap().as_clustering()
#     num_comm_ER_wa[i] = len(cl_wa)
```

In [77]:

```
# import igraph as ig
# import numpy as np
# import random
# import matplotlib.pyplot as plt

# # Set seeds
# random.seed(10)
# np.random.seed(10)

# # Number of trials
# n_trials = 10000

# # Degree sequence of your real London Underground network
# deg_london = ig_graph.degree()

# # Preallocate arrays
# num_comm_G_fg = np.zeros(n_trials)
# num_comm_G_lo = np.zeros(n_trials)
# num_comm_G_lp = np.zeros(n_trials)
# num_comm_G_eb = np.zeros(n_trials)
# num_comm_G_wa = np.zeros(n_trials)

# # Run community detection on randomized networks with degree pres
```

```
# i = 0
# while i < n_trials:
#     try:
#         # Generate a degree-preserving random graph (Viger–Latapy
#         G_rand = ig.Graph.Degree_Sequence(deg_london, method="vl"

#         # Fast Greedy
#         cl_fg = G_rand.community_fastgreedy().as_clustering()
#         num_comm_G_fg[i] = len(cl_fg)

#         # Louvain
#         cl_lo = G_rand.community_multilevel()
#         num_comm_G_lo[i] = len(cl_lo)

#         # Label Propagation (using Louvain membership as initial)
#         cl_lp = G_rand.community_label_propagation(initial=cl_lo.
#         num_comm_G_lp[i] = len(cl_lp)

#         # Edge Betweenness
#         cl_eb = G_rand.community_edge_betweenness(directed=False)
#         num_comm_G_eb[i] = len(cl_eb)

#         # Walktrap
#         cl_wa = G_rand.community_walktrap().as_clustering()
#         num_comm_G_wa[i] = len(cl_wa)

#         i += 1  # Only increment on success

#     except:
#         continue  # Skip and retry on failure
```